

Supplementary Material for “Innovation-Normalized Detection in Compound-Gaussian Clutter: Exact Laws, Conformal Thresholds and Certified Integration under Pulsed Interference”

M. U. Raza, S. Ahmed, and A. Ahmed

This supplement supports the paper’s conditional laws and measured calibration results. Numerical tables come from frozen saved outcomes; no new radar experiment was run. Protocols, saved summary outputs and plotting arrays are available in the accompanying repository (release `r21-trs-submission`, `research/r7c`); Section 18 gives their paths. Study labels identify internal hash-recorded protocols, not independent experiments or an external registry. Failed conditions and post-hoc choices remain explicit.

1 Data exposure

Table S1 states which data each stage used, so that the strength of each validation claim can be judged. Only the hold-out series, the 77 GHz data and NetRAD were not used while the methods were being developed.

Table S1: Data exposure by stage. “Seen” means that outcomes on these data were computed while methods, orders or comparisons could still change.

Data	Used for	Status of the reported results
IPIX, 14 sessions, rotation 1	Method development (R1–R7): method list, model orders, comparisons	Seen; not reported
IPIX, same sessions, rotations 2–3	Main study, detection, onset, base-lines, classical detectors, order-statistic scores (R7–R11)	Frozen-protocol evaluation, computed once per protocol; the recordings were seen in rotation 1, so this is not an untouched confirmation
IPIX, day transfer	Transfer and ACI analyses	Frozen-protocol evaluation on the same recordings
IPIX #269, #287	Hold-out	Hashed and unopened until the main analysis was complete; evaluated once (small test sets)
JKU 77 GHz, reference run	Second-radar study	Not used in development; frozen protocol; the general-form gain curve is post hoc
JKU 77 GHz, interference runs A and C	Real-interference test (R11); certified integration (R12)	R11: only chirp-hit statistics inspected before its protocol. R12: R11 outcomes on these runs had been seen
IPIX, rotations 2–3	Guarded scale, clip level, trimming, blanking (R14)	Frozen-protocol evaluation, computed once; same recordings as R7–R12
IPIX, primary target cells	Real-target study (R12)	Excluded from the R7–R11 clutter studies; a development check (V4, before R7) had run IN-ARCP on the primary cell and found it blind. Frozen protocol, evaluated once
NetRAD S-band, node 3, 14 recordings	Second-radar sea-clutter confirmation (R15)	Untouched: protocol frozen before any statistic of the samples was computed; evaluated once
JKU 77 GHz, pedestrian run (scenario A)	Real onsets of a walking person (R17)	Untouched: protocol frozen and pushed before the file was downloaded; evaluated once; descriptive (E1 not met)

2 Protocol deviations

- Main IPIX study: before any run, an encoding error in the protocol file was repaired byte-exactly and the hash rewritten; no outcome had been computed. A GPU backend was added for the likelihood evaluation of the noise-aware fit; it gives identical fitted parameters to the CPU path.
- Second radar: the descriptive hit rule summed range bins 20–247 instead of 20–250, because the fixed preprocessing removes bins above 247. The general-form gain curve was chosen after the pre-specified matched form failed to capture the magnitude of the measured gain, so it is post hoc.
- Order-statistic study (R11): the protocol’s rationale counted $j + 1$ target-corrupted history innovations at look j after onset; the correct count, j , was adopted after the data were seen. The data follow the corrected count, and the main text uses only the corrected statement.
- Final study (R12): before the real-data runs, the numerical search for the Fisher’s g threshold was re-bracketed; the law is unchanged.

3 Unmet expectations

Each expected value was written down before the outcome was computed. Of the 53 outcomes predicted for the last four studies, thirteen were not met and four, all from the pedestrian study, were left without a verdict.

- Pedestrian study (R17; E1 not met, E2–E5 without a verdict): the onsets were expected in at least eight distinct five-frame blocks and fell in seven, because the person was tracked for only about six seconds. Every other result is therefore descriptive; all lie on the side the protocol expected (Section 16).
- Final IPIX study (R12; 12 of 18 met): calibration of the saturating clipped statistic; the Tyler law (expected within twofold); the CA power detector’s white-noise law (expected off more than threefold); the cost of clipping (expected at most 1.5 dB); false alarms of non-coherent integration under real FMCW interference (expected at least fivefold in both runs); and the gain of AR gap reconstruction (expected at least 1 dB).
- Improvement study (R14; 12 of 15 met): the guard’s onset cost, the certificate at 10^{-3} and the cost of trimming (Section 12).
- NetRAD confirmation (R15; 12 of 15 met): calibration of six statistics at 10^{-4} , the pivotality ranking against CA16, and a rise of false alarms in segments with flagged pulses (Section 15).
- Main IPIX study ($m = 16$, $\alpha = 0.1$; radius relative to IN-ARCP): the noise-aware gain was expected to concentrate in noise-affected sessions. It did for NA-AR(1) (0.915 noisy against 0.989 clean) but not for NA-AR(2) and NA-AR(4) (0.871 and 0.866 noisy against 0.739 and 0.709 clean), whose gain comes mainly from the AR clutter dynamics (expectation Q1, mixed).
- Onset study: the gain of IN-ARCP over the long-window power detector was expected to shrink below 3 dB by an 8-pulse dwell. It persisted: 8.4 dB [5.2, 11.1] at $P_d = 0.5$ (expectation O3, refuted).
- 77 GHz radar: the noise-aware scores were expected to be flatter than IN-ARCP across CNR classes. The AR(4) version was (mean largest class deviation 0.055 against 0.066); the AR(1) version was not (0.067; expectation J2, mixed).
- Classical detectors (R10). The frozen `r10/r10.summary.txt` prints expectation R1 as FAILED. It is met in substance. For a target that rises over L pulses, reaching full amplitude at the first look, IN-ARCP’s first-look gain over the long-window power detector (CAloc in the paper) falls from 9.86 to 8.65 to 4.65 dB for $L = 1, 4, 8$. At $L = 16$, CAloc reaches $P_d = 0.5$ at 9.1 dB while IN-ARCP stays below it up to 25 dB, so the gain is below -15.9 dB, well under the 5 dB bound. PAMF-H was expected to need at least 3 dB less SCR than integrated IN-ARCP over an 8-pulse dwell at $P_d = 0.5$; it needed 1.0 dB [0.3, 1.7] less (4.0 dB at $P_d = 0.9$). Gated ACI was expected to keep the clean false-alarm rate within $[0.8, 1.2]\alpha$ at 5% target contamination; it gave 0.55α ($\gamma = 0.02$) and 0.68α ($\gamma = 0.005$).
- Order-statistic study (R11; 2 of 11 met): quintile false-alarm spread of the median OS score at most 2.0 (2.06); its cost for abrupt targets at most 1.5 dB (1.99 dB [0.89, 3.39]; Corollary 3 gives 0.69 dB); a first-look gain of at least 3 dB over IN-ARCP for 8-pulse ramps (1.46 dB); the integrated OS score closing half of integrated IN-ARCP’s gap to PAMF-H at $P_d = 0.9$ (it was 2.5 dB worse); OS-law coverage error at most 0.015 (0.021); on the dense real-interference run, the OS score halving IN-ARCP’s masking (drops 0.39 and 0.35), zeroing cutting the hit-pulse false-alarm rate by half (39%), and the OS score after zeroing within 0.5 dB of IN-ARCP (0.71 dB worse); on the reference run, IN-ARCP beating CA16 (CA16 was 0.78 dB better, the loss Proposition 2 predicts for $c < 1$).

4 Proof of Proposition 2 and the latent-texture Mondrian result

Equation, proposition and theorem numbers without the prefix S refer to the paper.

Proposition 2. If $e \sim \mathcal{CN}(0, v)$ and a Swerling-1 target of power SP is added, a threshold test on $|e|^2/v$ at false-alarm probability α detects with probability $\alpha^{1/(1+SP/v)}$, while the power-only test on $|Y|^2/P$ detects with $\alpha^{1/(1+S)}$; equal P_d therefore requires SCRs in the ratio $G = P/v$. For the center rH_m under (1) with AR(1) speckle, $v = \nu\{c(1 + |r|^2 - 2\Re \bar{r}\rho) + 1 + |r|^2\}$, which gives G_{IN} . For the optimal predictor from the infinite past, v is the one-step prediction-error variance of the AR(1)-plus-white-noise process, $\exp\{\int \log f\}$ by the Kolmogorov–Szegő formula [1]. With $f(\omega) = \nu\{c(1 - |\rho|^2) + |1 - \rho e^{-j\omega}|^2 / |1 - \rho e^{-j\omega}|^2\}$, the denominator contributes zero, and the numerator factorizes as $\nu V |1 - \zeta e^{-j\omega}|^2$ with $V(1 + |\zeta|^2) = \xi$, $V\zeta = \rho$ and $|\zeta| < 1$, which gives $v = \nu V$ and G_{opt} . Because the optimal predictor’s error variance is at most that of any linear predictor, including 0 and rH_m , $G_{\text{opt}} \geq \max\{1, G_{\text{IN}}\}$.

Latent-texture Mondrian calibration. Mondrian (group-conditional) conformal prediction calibrates separately within bins of an observed statistic and is valid conditional on the observed bin; Dewolf *et al.* report that estimated taxonomies mix up the true classes [2]. For a latent texture the consequence can be stated exactly.

Proposition S1. Let $\nu = 0$ and $X = \sigma Z$, with $\sigma > 0$ distributed as G , non-degenerate and independent of $Z \sim \mathcal{CN}(0, R)$. Let $S(Z)$ be a scale-invariant score with a continuous distribution function F_S that is strictly increasing on its support, let $T(H) > 0$ be continuous and homogeneous of degree one, and calibrate within fixed bins $B_b = [e_{b-1}, e_b)$, $0 = e_0 < \dots < e_K = \infty$, $K \geq 2$, of $T(H)$ with population quantiles q_b . Let $\mathcal{C}(\sigma)$ be the coverage conditional on the texture. (i) If $S(Z)$ is independent of $T(Z_H)$, then $\mathcal{C}(\sigma) \equiv 1 - \alpha$. (ii) If $S(Z)$ given $T(Z_H) = t$ is strictly stochastically decreasing in t , then $\lim_{\sigma \rightarrow \infty} \mathcal{C}(\sigma) = F_S(q_K) < 1 - \alpha < F_S(q_1) = \lim_{\sigma \rightarrow 0} \mathcal{C}(\sigma)$, whereas the unbinned threshold gives a constant $\mathcal{C}(\sigma)$. Condition (ii) holds exactly for IN-ARCP at the true coefficient with $T = s_\rho$.

Proof. Write $t = T(Z_H)$, so that the observed statistic is $T(X_H) = \sigma t$ and $S(X) = S(Z)$. For bin b let $w_b(t) = \mathbb{P}\{\sigma t \in B_b\}$. Since σ is independent of Z , for every x

$$\mathbb{P}\{S \leq x, \sigma t \in B_b\} = \mathbb{E}[\varphi_x(t) w_b(t)], \quad (\text{S1})$$

where $\varphi_x(t) = \mathbb{P}\{S \leq x \mid t\}$. (i) If S is independent of t , then $\varphi_x \equiv F_S(x)$, every bin-conditional law of S equals F_S , all q_b equal $F_S^{-1}(1 - \alpha)$ and $\mathcal{C}(\sigma) = 1 - \alpha$.

(ii) Let $x_0 = F_S^{-1}(1 - \alpha)$. For the top bin, $w_K(t) = \mathbb{P}\{\sigma \geq e_{K-1}/t\}$ is nondecreasing in t . Because T is continuous, positive and homogeneous and Z_H has full support, t has full support on $(0, \infty)$; as e_{K-1}/t then ranges over $(0, \infty)$ and G is non-degenerate, $w_K(t)$ is not almost surely constant. By assumption φ_{x_0} is strictly increasing. Let t' be an independent copy of t , $D_\varphi = \varphi_{x_0}(t) - \varphi_{x_0}(t')$ and $\Delta_w = w_K(t) - w_K(t')$. Then

$$\text{Cov}\{\varphi_{x_0}(t), w_K(t)\} = \frac{1}{2} \mathbb{E}[D_\varphi \Delta_w] > 0,$$

because $D_\varphi \Delta_w \geq 0$, with strict inequality on the event $\{w_K(t) \neq w_K(t')\}$, which has positive probability. By (S1), $\mathbb{P}\{S \leq x_0 \mid \sigma t \in B_K\} = \mathbb{E}[\varphi_{x_0} w_K] / \mathbb{E}[w_K] > \mathbb{E}[\varphi_{x_0}] = 1 - \alpha$. The bin-conditional distribution function of S is continuous, so there is $x_1 < x_0$ at which it still exceeds $1 - \alpha$; hence $q_K \leq x_1 < x_0$ and, F_S being strictly increasing, $F_S(q_K) < 1 - \alpha$. For the bottom bin, $w_1(t) = \mathbb{P}\{\sigma < e_1/t\}$ is nonincreasing, the covariance is negative, and the same steps give $q_1 > x_0$ and $F_S(q_1) > 1 - \alpha$.

The coverage at texture σ is $\mathcal{C}(\sigma) = \sum_b \mathbb{P}\{S \leq q_b, \sigma t \in B_b\}$. As $\sigma \rightarrow \infty$, $\sigma t \rightarrow \infty$ almost surely, so $\mathbf{1}\{\sigma t \in B_K\} \rightarrow 1$ and, by dominated convergence, $\mathcal{C}(\sigma) \rightarrow F_S(q_K)$; as $\sigma \rightarrow 0$, $\mathcal{C}(\sigma) \rightarrow F_S(q_1)$. An unbinned threshold q gives $\mathcal{C}(\sigma) = F_S(q)$ for every σ .

Finally, for IN-ARCP at $r = \rho$ and $T = s_\rho$, $S = |E|/t$ with the innovation E independent of Z_H , so $\varphi_x(t) = \mathbb{P}\{|E| \leq xt\}$ is strictly increasing in t , which is condition (ii). $\blacksquare$

5 Zero-noise limit of the noise-aware score

Proposition S2. Let R be AR(1) with coefficient ρ and fix a nonzero history h . As $\nu \downarrow 0$, $\nu \hat{c}(h) \rightarrow h^H R_H^{-1} h/m$, $\mu(h) \rightarrow \rho h_m$ and $s(h) \rightarrow s_\rho(h)$, so the NA score converges to the IN-ARCP score at $r = \rho$.

Proof sketch. With $\gamma = \nu c$ the history log-likelihood converges locally uniformly on $\gamma > 0$ to $-m \log \gamma - \gamma^{-1} h^H R_H^{-1} h$, whose unique maximizer is $h^H R_H^{-1} h/m$; coercivity at 0 and ∞ gives convergence of the maximizers. The Markov property gives $\mu \rightarrow \rho h_m$, and $(1 - |\rho|^2) R_H^{-1} = W_\rho^H W_\rho$, with W_ρ the bidiagonal whitening matrix, identifies the limiting predictive variance with $s_\rho(h)^2$.

6 Proofs of the whitened-innovation laws

Throughout, $z \sim \mathcal{CN}(0, I)$ denotes whitened clutter innovations (unit variance after dividing by σ_e), $g \sim \mathcal{CN}(0, 1)$ the Swerling-1 amplitude, and u the whitened target signature, so that the whitened episode is $z + gu \sim \mathcal{CN}(0, I + uu^H)$. E , $E_{\pm}$ are standard exponentials and Γ_k a Gamma($k, 1$) variable, all independent.

6.1 Proposition (post-onset rank-one law)

Write the detection event as $F > 0$ with $F = x^H M x$, $x = z + gu$ and $M = \text{diag}(-\beta I_m, 1)$, $\beta = q^2/m$. As in Appendix A of the paper, $F \stackrel{d}{=} \sum_k \lambda_k E_k$ with λ_k the eigenvalues of $C^{1/2} M C^{1/2}$, $C = I + uu^H$, which are those of $MC = M + (Mu)u^H$. For a rank-one update of a diagonal matrix,

$$\det(\lambda I - M - Muu^H) = \det(\lambda I - M) \left(1 - \sum_i \frac{M_{ii}|u_i|^2}{\lambda - M_{ii}}\right) = (\lambda - 1)(\lambda + \beta)^m \left(1 - \frac{\mathcal{E}_0}{\lambda - 1} + \frac{\beta \mathcal{E}_H}{\lambda + \beta}\right),$$

with $\mathcal{E}_0 = |u_Y|^2$ and $\mathcal{E}_H = \|u_H\|^2$. Hence $-\beta$ is an eigenvalue of multiplicity $m - 1$ and the other two solve $(\lambda - 1)(\lambda + \beta) - \mathcal{E}_0(\lambda + \beta) + \beta \mathcal{E}_H(\lambda - 1) = 0$, which is the quadratic of the proposition. Their product is $-\beta(1 + \mathcal{E}_0 + \mathcal{E}_H) < 0$, so $\lambda_+ > 0 > \lambda_-$. Then

$$\mathbb{P}\{F > 0\} = \mathbb{P}\{\lambda_+ E_+ > |\lambda_-| E_- + \beta \Gamma_{m-1}\} = \mathbb{E} e^{-(|\lambda_-| E_- + \beta \Gamma_{m-1})/\lambda_+} = \frac{\lambda_+}{\lambda_+ + |\lambda_-|} \left(1 + \frac{\beta}{\lambda_+}\right)^{-(m-1)}.$$

For $\mathcal{E}_H = 0$ the roots are $1 + \mathcal{E}_0$ and $-\beta$, and the expression reduces to $(1 + \beta/(1 + \mathcal{E}_0))^{-m}$. The signature of a persistent target follows from the whitened amplitudes in (4): the onset innovation carries S_w , every later one $S_w|d|^2$.

6.2 Corollary (visibility horizon)

With $\mathcal{E}_0 = S_w D$ and $\mathcal{E}_H = S_w(1 + (\ell - 1)D)$, $D = |d(\omega)|^2$, the linear coefficient is $B = 1 - \beta + S_w \chi$, $\chi = D - \beta\{1 + (\ell - 1)D\}$, and $\lambda_{\pm} = \{B \pm (B^2 + 4\beta(1 + \mathcal{E}_0 + \mathcal{E}_H))^{1/2}\}/2$ with $1 + \mathcal{E}_0 + \mathcal{E}_H = O(S_w)$. If $\chi > 0$, then $\lambda_+ \sim S_w \chi \rightarrow \infty$ while λ_- stays bounded, so $P_d \rightarrow 1$. If $\chi < 0$, then $\lambda_+ \rightarrow \beta(1 + \ell D)/|\chi|$ stays bounded while $\lambda_- \sim S_w \chi \rightarrow -\infty$, so $\lambda_+ / (\lambda_+ - \lambda_-) \rightarrow 0$ and $P_d \rightarrow 0$. The sign change $\chi = 0$ is $\ell = 1 + 1/\beta - 1/D$. At equality, $\lambda_{\pm} \sim \pm \sqrt{\beta S_w(1 + \ell D)}$ and $P_d \rightarrow 1/2$. The horizon describes the post-onset interval in which the onset innovation remains in the history, not look zero. With a guard, only the history energy is delayed. Within protected looks $1, \dots, \Delta$, $P_d \rightarrow 1$ as $S \rightarrow \infty$ for $D > 0$. The critical value $\Delta + \ell^*$ applies beyond those looks; if $\ell^* < 1$, loss first occurs at $\Delta + 1$ (at $\ell^* = 1$ that look tends to $1/2$).

6.3 Order-statistic scale

Conditional on $G = |g|^2$, the history powers $|z_j + gu_j|^2$ are independent, exponential for clean innovations and non-central (with $2|z_j + gu_j|^2 \sim \chi_2^2(2G|u_j|^2)$) for contaminated ones. The distribution function of their k -th smallest value $X_{(k)}$ is the Poisson-binomial probability that at least k are below x , and $P_d = \mathbb{E}_G \int \mathbb{P}\{|z_Y + gu_Y|^2 > q^2 x \mid G\} dF_{X_{(k)}|G}(x)$. The code evaluates this by midpoint quadrature in the distribution function of G and a dynamic program for the Poisson-binomial law. If $\ell \leq m - k$ contaminated powers diverge, $X_{(k)}$ is the k -th smallest of the $m - \ell$ clean ones, which is independent of z_Y , and the OS-CFAR law with $m - \ell$ cells gives (7). If $\ell > m - k$, let $j = k - (m - \ell)$ and let $a_{(j)}$ be the j -th smallest of $\{1, D, \dots, D\}$ (ℓ entries), $D = |d|^2$. The tested-power/scale ratio converges almost surely to $D/a_{(j)}$. Hence $P_d \rightarrow 0$ when $q^2 > D/a_{(j)}$, in particular when $q^2 > \max\{1, D\}$, as for the reported $k = 8, 12$ settings at $\alpha = 0.01$. Outside this condition self-masking is not assured.

6.4 Dwell integration

Under H_0 the event is $\Gamma_K > \tau \Gamma_m$, and conditioning on Γ_m with the Poisson form of the Gamma tail gives $\sum_{i < K} \mathbb{E}[e^{-\tau \Gamma_m} (\tau \Gamma_m)^i / i!] = \sum_{i < K} \binom{m+i-1}{i} \tau^i (1 + \tau)^{-(m+i)}$. With a Swerling-1 target the dwell form has one eigenvalue $\delta = 1 + \mathcal{E}_D$ and $K - 1$ unit ones. For $v > 0$,

$$\mathbb{P}\{\delta E + \Gamma_{K-1} > v\} = \mathbb{P}\{\Gamma_{K-1} > v\} + e^{-v/\delta} (1 - 1/\delta)^{-(K-1)} \mathbb{P}\{\Gamma_{K-1} \leq v(1 - 1/\delta)\}.$$

Averaging over $v = \tau \Gamma_m$ uses $\mathbb{E} e^{-s \Gamma_m} = (1 + s)^{-m}$ and $\mathbb{E}[e^{-s \Gamma_m} \mathbb{P}\{\Gamma_{K-1} > \tau' \Gamma_m \mid \Gamma_m\}] = \sum_{i \leq K-2} \binom{m+i-1}{i} \tau'^i (1 + s + \tau')^{-(m+i)}$ with $s + \tau' = \tau$, which gives (9) for $\delta > 1$. At $\delta = 1$ its continuous limit is (8).

6.5 Self-normalized whitened Doppler

With a known on-grid Doppler, the whitened dwell vector splits into its DFT coefficient on the target bin, $|\cdot|^2 \sim (1 + NS_w|d|^2)E$, and the orthogonal remainder Γ_{N-1} . The statistic $X/(X + \Gamma_{N-1})$ exceeds t iff $X(1-t) > t\Gamma_{N-1}$, which has probability $\{1 + t/((1-t)(1 + NS_w|d|^2))\}^{-(N-1)}$; at $S_w = 0$ this is $(1-t)^{N-1}$. For the maximum over the N bins, let I_j be independent exponentials with means μ_j and J a set of bins with $|J|t < 1$. Substituting $x_j = tT + y_j$ for $j \in J$, where $T = \sum_i I_i$, the map has Jacobian $1/(1 - |J|t)$ and turns the exponent into a sum of independent exponentials, so

$$\mathbb{P}\{I_j > tT \ \forall j \in J\} = \frac{1}{1 - |J|t} \prod_{i=1}^N \frac{1}{1 + \theta_J \mu_i}, \quad \theta_J = \frac{t \sum_{j \in J} \mu_j^{-1}}{1 - |J|t},$$

and zero if $|J|t \geq 1$. With all $\mu_i = 1$ this is $(1 - |J|t)^{N-1}$, and inclusion-exclusion gives Fisher's law $\sum_r (-1)^{r-1} \binom{N}{r} (1 - rt)^{N-1}$ (R. A. Fisher, Proc. R. Soc. Lond. A 125, 1929). With one bin of mean $1 + NS_w|d|^2$, inclusion-exclusion over sets that do or do not contain it gives P_d (`laws.pd.snd.max`).

6.6 Theorem (certified integration)

Domination. Let $\mathcal{J}_H$ and $\mathcal{J}_D$ be the corrupted history and dwell innovations. The k_H -th smallest of the history powers is a nondecreasing function of each power, so over all values of the corrupted entries it is smallest when they are zero; hence $s_{k_H}^2 \geq \underline{s}^2$, the value computed with those entries set to zero. If $\underline{s}^2 = 0$, define the bound as $W = K\kappa$ (binary: K); the bound then holds trivially. For an actual zero scale, choose ratios so each clipped term remains in $[0, \kappa]$ (binary: $[0, 1]$); the saturated bound still dominates. Otherwise every uncorrupted dwell term is a clean power divided by $s_{k_H}^2 \geq \underline{s}^2$, and every clipped term is at most κ . Therefore $T \leq W$ for all interference values. For binary integration, an uncorrupted exceedance of the actual statistic is an exceedance at $\underline{s}^2$, and a corrupted term counts at most one. Adding elements to $\mathcal{J}$ can only lower $\underline{s}^2$ and replace terms by their maximum, so W is nondecreasing in $\mathcal{J}$. *Calibration.* Draw the calibration masks i.i.d. from $\mathcal{D}$, independently of all clean episodes and of the test hit set. Let the test hit set $\mathcal{J}$ be independent of all clean episodes and couple it, independently of these episodes and the calibration masks, to $\tilde{\mathcal{J}} \sim \mathcal{D}$ with $\mathcal{J} \subseteq \tilde{\mathcal{J}}$ (a coupling that exists when $\mathcal{D}$ dominates the law of $\mathcal{J}$ in the inclusion order, by Strassen's theorem; a bound on the hit rate alone does not suffice). Then $T \leq W(X, \mathcal{J}) \leq W(X, \tilde{\mathcal{J}})$. The pairs $(X_i, \mathcal{J}_i)$ and $(X, \tilde{\mathcal{J}})$ are exchangeable, so the split-conformal rank argument gives $\mathbb{P}\{W(X, \tilde{\mathcal{J}}) > \hat{q}\} \leq \alpha$. No property of the interference values is used, so the bound holds for adversarial amplitudes chosen after seeing the clutter, as long as the hit positions are independent of the clutter. *Trimmed sums.* If a corrupted dwell term can take any value, the trimmed sum over the $K - t$ smallest terms is unbounded once more than t dwell terms are corrupted, which a Bernoulli hit model allows with positive probability. If the history lower bound is positive almost surely, this is exactly the source of infinite dwell terms. The population quantile is infinite when their probability exceeds α ; a finite-sample threshold is infinite only when enough sampled bounds are infinite. History-scale collapse can also make the trimmed bound infinite.

6.7 Conformal calibration size and loss

For continuous i.i.d. scores with $k_\alpha \leq n$, the k_α -th order statistic of n calibration scores, mapped through the score distribution function, is $\text{Beta}(k_\alpha, n + 1 - k_\alpha)$; the conditional false-alarm probability is one minus it, hence $\text{Beta}(n + 1 - k_\alpha, k_\alpha)$. Because k_α is a ceiling, $\mathbb{P}\{P_{\text{fa}} > 2\alpha\}$ is not monotone in n ; For $\alpha = 10^{-2}, 10^{-3}, 10^{-4}$, an exhaustive scan gives the smallest n and the n beyond which the condition always holds. The finite scan is completed by a tail bound: writing $L = \lfloor (n+1)\alpha \rfloor$, $\mathbb{P}\{P_{\text{fa}} > 2\alpha\} = \mathbb{P}\{\text{Bin}(n, 2\alpha) \leq L - 1\}$. Since $L - 1 \leq n\alpha$, Chernoff gives an upper bound $e^{-n\alpha/4} < 0.05$ for $n \geq 12/\alpha$. Thus only the ranks below $12/\alpha$ require enumeration. With ties the rank argument still gives $\mathbb{P}\{T > \hat{q}\} \leq \alpha$, but the Beta law no longer applies. For $P_d = P_{\text{fa}}^\gamma$, $\mathbb{E}U^\gamma = B(n + 1 - k_\alpha + \gamma, k_\alpha)/B(n + 1 - k_\alpha, k_\alpha)$, with B the Beta function.

7 Numerical verification and frozen-output corrections

The saved `r12/verify_laws_output.txt` contains 400,000-replication Monte Carlo checks. Its 65 primary checks are the rows with z -scores excluding bracketed labels: two compound-Gaussian tests of the Gaussian Kraut-Scharf law and four asymptotic fixed-point-law checks. Every primary check lies within three standard errors (maximum $|z| = 2.9125$, printed as 2.91); the fixed-point checks have $|z| < 1.6$. Beta and certificate rows report distribution agreement or violation rates, not z -scores. The compound-Gaussian Kraut-Scharf tests fail 2.8–3.4-fold at 10^{-2} ; this failure remains part of the evidence.

The independent `r14/theory_check_independent.py` imports no implementation of the manuscript laws. Its saved output checks repeated eigenvalues, singular scales, the post-onset law, horizon, tested strong-target OS regime, integration, known-bin P-ANMF, Beta calibration, and certificates under strong, matched and cancelling amplitudes. Bursts

calibrated as independent hits violate design (clipped $P_{\text{fa}} = 0.0114$, binary 0.0192 at design 0.01). Matched burst certificates record no exceedance and lose sensitivity. The separate guarded-AR check has maximum $|z| = 2.56$ with 400,000 episodes per point. Protected looks use unchanged tested energy and zero target energy in the history; only the contaminated interval shifts by the guard.

Two corrections accompany the unchanged full logs. V8 calibration sizes in `verify_laws_output.txt` came from a faulty search. `r12/calibration_size.txt` supersedes them: first acceptable sizes 149, 1,497 and 14,978; permanent cutoffs 313, 3,146 and 31,477 at $\alpha = 10^{-2}, 10^{-3}, 10^{-4}$. The proof supplies the tail completion. The independent log's $n = 23,000$ line checks a formula, not the actual set size. Actual per-look calibration sets contain 17,164–23,130 episodes; their i.i.d. Beta-tail probabilities are about 0.02–0.09. Non-overlap does not imply independence. The R10 ramp log prints R1 as FAILED because a numerical crossing is unavailable when IN-ARCP never reaches $P_d = 0.5$ by 25 dB; the recorded curves meet that expectation in substance, as Section 3 explains.

7.1 Closed forms against Monte Carlo: all six panels

Figure S1 checks the post-onset, horizon, order-statistic, integration, Doppler and certificate calculations.

8 False-alarm calibration and pivotality

Tables S2 and S3 preserve every Part-L ratio and cluster interval, including the residual bootstrap. They are measured pooled rates, not guarantees of exchangeability or equal rates in individual cells. IPIX intervals resample six days; NetRAD intervals resample fourteen recordings. These intervals are approximate. Saturation and ties can make clipped thresholds conservative.

8.1 Pivotality across clutter power, all thirteen statistics

Table S4 gives the texture-quintile spread, the ratio of the largest to the smallest measured false-alarm rate over quintiles of local clutter power, for every statistic of Table I under both threshold rules. The spread is a pre-specified outcome of the frozen protocol; the two readings of it below are exploratory.

The remedy recommended for persistent targets has a strongly texture-dependent rate. Proposition 3 rules out history normalization for a target present throughout the history, and the paper recommends the self-normalized whitened Doppler statistic instead. Its conformal rate is compliant when pooled (1.04 at 10^{-2} , 1.47 at 10^{-4}) but varies by a factor of 23 across texture quintiles at 10^{-4} , against 2.7 for IN-ARCP. It carries the widest spread in the table at 10^{-2} , though not at the two lower design rates, where the saturating clipped integrator is wider still. The statistic is scale-invariant and therefore texture-pivotal in pure compound-Gaussian clutter. Additive noise can break this invariance (Corollary 1); residual correlation and model mismatch can also contribute. The quintile spreads do not isolate the cause. The dwell stays in one range cell.

The model-based laws fail in two distinct ways. Among the rules that exceed twice design at 10^{-4} , the order-statistic and maximum-DFT laws carry the largest texture spreads (4.0 and 4.5 at 10^{-3} , against 1.4 for the CA law of IN-ARCP), so their rate depends strongly on clutter power. The summary prints only the max/min ratio, but the saved outcome arrays retain all five quintile exceedance counts and denominators. No direction or cause follows from a spread alone. Three other rules behave differently: the clipped integrator's simulated threshold fails worse still in the pooled rate (17.19 at 10^{-3} , 59.89 at 10^{-4}) and the two ANMF laws less badly (3.57 and 2.66 at 10^{-3}), but all three almost evenly across quintiles (1.3, 2.0 and 2.2). For the Kraut–Scharf law that is expected, since the paper's simulation shows it already failing 2.8–3.4-fold at 10^{-2} in compound-Gaussian clutter, where the real failure is smaller. Attributing the difference to tail weight would require an analysis beyond these spread summaries; it is not established here.

9 Coverage, width and transfer checks

9.1 Synthetic verification

Figure S2 compares exact and simulated clutter-to-noise-ratio-conditional coverage at nominal 0.90.

9.2 Texture-conditional coverage and disc widths on IPIX

Figure S3 compares texture-conditional coverage; the following table reports coverage and disc widths.

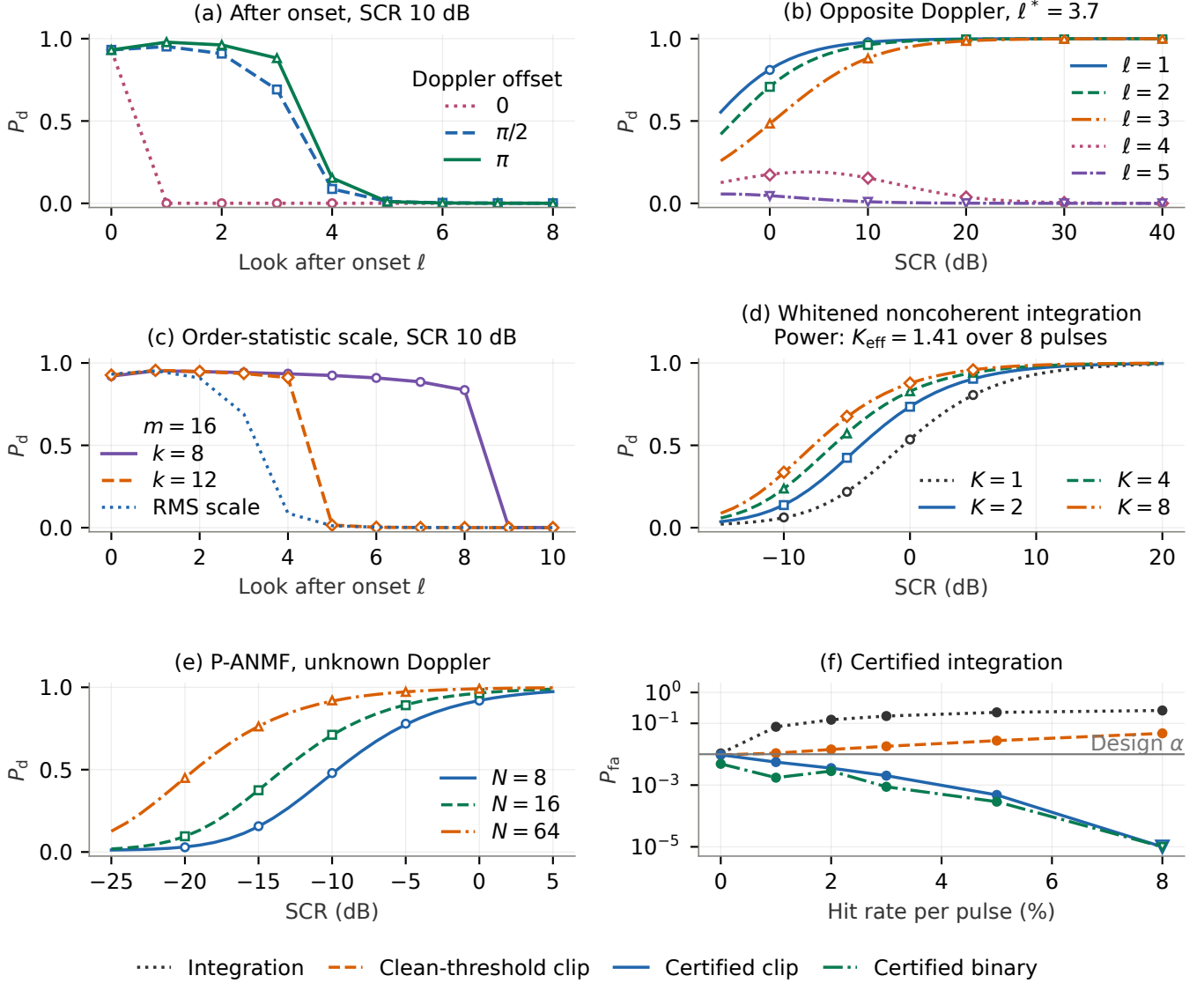


Figure S1: Closed forms (lines) against Monte Carlo (markers; 10^5 episodes per point in (a), (c) and (f), 5×10^4 in (b) and (d), 10^4 to 2.5×10^4 in (e)) at the true coefficient, AR(1) compound-Gaussian clutter with $\rho = 0.93e^{-0.2j}$, lognormal texture, $m = 16$, $P_{fa} = 0.01$. Panels (a)–(c) and (f) are those of Fig. 2 of the paper, where (f) appears as its panel (d) and (c) here adds the RMS-scale law for reference. The two panels shown only here are (d), the integration law (9), and (e), P-ANMF with unknown Doppler. Doppler offsets are relative to clutter. In (f), interference is 30 dB above local power; curves join Monte Carlo estimates. Open downward triangles mark no false alarm in 10^5 episodes and are drawn at the resolution 10^{-5} .

Table S2: IPIX pooled P_{fa}/α [95% day-cluster interval]. Bold point estimates lie outside [0.5, 2]. Model rules are those named in the paper’s low-rate table.

Statistic	Conformal	Model-based rule
$\alpha = 10^{-2}$		
ANMF-Tyler	1.04 [0.98, 1.07]	1.62 [1.39, 1.82]
ANMF-SCM	1.06 [0.99, 1.12]	2.02 [1.69, 2.28]
Power CA	1.00 [0.99, 1.01]	0.80 [0.71, 0.85]
Clipped, OS scale	1.05 [0.99, 1.08]	4.84 [3.72, 6.72]
Integration, AR(4)	1.02 [0.98, 1.07]	2.02 [1.73, 2.27]
IN-ARCP	1.01 [0.99, 1.02]	0.96 [0.92, 1.02]
IN-AR(4)	1.00 [0.99, 1.01]	1.03 [1.01, 1.05]
NA-AR(4)	1.00 [0.95, 1.08]	1.56 [1.40, 1.70]
OS scale, $k = 8$	1.04 [1.01, 1.08]	2.15 [1.06, 3.57]
Power OS	1.01 [0.96, 1.04]	2.07 [1.47, 2.55]
P-ANMF, maximum	1.04 [0.95, 1.11]	4.36 [3.58, 5.29]
P-ANMF, one bin	1.10 [1.01, 1.19]	1.78 [0.73, 2.61]
PAMF-H, one Doppler	0.99 [0.96, 1.03]	1.97 [1.06, 2.70]
IN-ARCP, residual bootstrap	—	1.63 [1.47, 1.75]
$\alpha = 10^{-3}$		
ANMF-Tyler	1.05 [0.96, 1.12]	2.66 [1.76, 3.48]
ANMF-SCM	1.04 [0.94, 1.14]	3.57 [2.38, 4.54]
Power CA	1.00 [0.94, 1.06]	0.56 [0.41, 0.63]
Clipped, OS scale	0.67 [0.34, 0.95]	17.19 [11.68, 25.51]
Integration, AR(4)	1.05 [0.86, 1.20]	4.35 [3.16, 5.33]
IN-ARCP	1.03 [1.01, 1.04]	1.02 [0.91, 1.12]
IN-AR(4)	1.02 [0.96, 1.08]	1.19 [1.01, 1.31]
NA-AR(4)	1.05 [1.00, 1.08]	2.79 [2.38, 3.24]
OS scale, $k = 8$	1.02 [0.89, 1.13]	8.34 [2.00, 15.87]
Power OS	0.98 [0.91, 1.02]	5.56 [3.15, 7.36]
P-ANMF, maximum	1.04 [0.90, 1.16]	13.06 [10.39, 17.19]
P-ANMF, one bin	1.14 [0.95, 1.26]	3.05 [1.01, 5.06]
PAMF-H, one Doppler	1.00 [0.89, 1.08]	3.66 [1.66, 5.60]
IN-ARCP, residual bootstrap	—	2.43 [2.15, 2.67]
$\alpha = 10^{-4}$		
ANMF-Tyler	0.82 [0.67, 0.95]	6.01 [2.87, 9.15]
ANMF-SCM	0.85 [0.75, 1.01]	8.07 [3.98, 12.11]
Power CA	0.75 [0.68, 0.90]	0.49 [0.26, 0.70]
Clipped, OS scale	0.03 [0.00, 0.11]	59.89 [40.97, 85.35]
Integration, AR(4)	1.02 [0.59, 1.31]	14.25 [7.84, 19.14]
IN-ARCP	0.73 [0.54, 0.99]	1.92 [1.26, 2.61]
IN-AR(4)	0.86 [0.64, 1.11]	2.73 [1.48, 3.67]
NA-AR(4)	1.04 [0.84, 1.19]	7.46 [4.55, 9.73]
OS scale, $k = 8$	0.92 [0.76, 1.14]	45.91 [7.96, 90.46]
Power OS	0.75 [0.57, 1.07]	18.33 [8.92, 25.59]
P-ANMF, maximum	1.47 [0.75, 2.57]	49.61 [35.38, 73.34]
P-ANMF, one bin	1.10 [0.79, 1.31]	5.47 [1.35, 9.99]
PAMF-H, one Doppler	0.89 [0.55, 1.22]	9.88 [4.13, 15.83]
IN-ARCP, residual bootstrap	—	4.27 [3.44, 4.74]

Table S3: NetRAD pooled P_{fa}/α [95% recording-cluster interval]. Bold point estimates lie outside [0.5, 2]. Model rules are those named in the paper’s low-rate table.

Statistic	Conformal	Model-based rule
$\alpha = 10^{-2}$		
ANMF-Tyler	1.01 [0.98, 1.03]	2.01 [1.97, 2.04]
ANMF-SCM	1.00 [0.98, 1.02]	2.27 [2.20, 2.33]
Power CA	1.01 [0.99, 1.03]	0.32 [0.27, 0.36]
Clipped, OS scale	1.05 [1.02, 1.10]	2.04 [1.89, 2.23]
Integration, AR(4)	1.06 [1.00, 1.15]	1.44 [1.36, 1.56]
IN-ARCP	1.04 [0.99, 1.09]	0.80 [0.63, 0.94]
IN-AR(4)	1.03 [1.00, 1.06]	0.98 [0.95, 1.02]
NA-AR(4)	1.08 [1.00, 1.19]	1.90 [1.75, 2.03]
OS scale, $k = 8$	1.05 [0.99, 1.14]	0.91 [0.73, 1.07]
Power OS	1.01 [0.99, 1.02]	0.65 [0.60, 0.70]
P-ANMF, maximum	1.04 [0.97, 1.12]	2.46 [2.29, 2.66]
P-ANMF, one bin	0.98 [0.90, 1.09]	2.28 [1.89, 2.66]
PAMF-H, one Doppler	1.03 [0.97, 1.10]	2.38 [2.05, 2.72]
IN-ARCP, residual bootstrap	—	0.82 [0.55, 1.05]
$\alpha = 10^{-3}$		
ANMF-Tyler	1.01 [0.94, 1.12]	3.00 [2.82, 3.21]
ANMF-SCM	1.02 [0.95, 1.11]	3.52 [3.29, 3.75]
Power CA	1.07 [1.03, 1.12]	0.18 [0.12, 0.23]
Clipped, OS scale	1.13 [1.05, 1.22]	4.40 [3.63, 5.45]
Integration, AR(4)	1.36 [0.96, 1.99]	2.42 [1.88, 3.22]
IN-ARCP	1.12 [0.99, 1.26]	0.90 [0.63, 1.13]
IN-AR(4)	1.08 [0.97, 1.20]	1.06 [0.95, 1.24]
NA-AR(4)	1.36 [1.02, 1.75]	4.23 [3.53, 4.85]
OS scale, $k = 8$	1.22 [0.98, 1.50]	1.17 [0.80, 1.53]
Power OS	1.03 [1.01, 1.06]	1.09 [0.91, 1.28]
P-ANMF, maximum	1.16 [0.85, 1.61]	4.33 [3.67, 5.15]
P-ANMF, one bin	1.15 [0.79, 1.75]	3.50 [2.77, 4.24]
PAMF-H, one Doppler	1.15 [0.98, 1.42]	4.09 [3.32, 5.08]
IN-ARCP, residual bootstrap	—	0.75 [0.32, 1.16]
$\alpha = 10^{-4}$		
ANMF-Tyler	1.18 [0.91, 1.63]	4.73 [4.08, 5.81]
ANMF-SCM	1.10 [0.91, 1.41]	5.64 [4.90, 6.63]
Power CA	1.18 [1.04, 1.33]	0.17 [0.07, 0.25]
Clipped, OS scale	0.39 [0.13, 0.65]	11.86 [8.29, 16.54]
Integration, AR(4)	4.26 [1.46, 9.24]	7.78 [3.95, 13.54]
IN-ARCP	1.30 [0.94, 1.72]	1.70 [0.91, 2.66]
IN-AR(4)	1.31 [1.02, 1.60]	1.85 [1.05, 3.07]
NA-AR(4)	2.08 [1.14, 3.34]	12.69 [9.13, 16.26]
OS scale, $k = 8$	1.38 [0.82, 2.05]	2.76 [1.25, 4.85]
Power OS	1.02 [0.91, 1.10]	2.21 [1.69, 2.78]
P-ANMF, maximum	2.33 [0.85, 5.19]	9.25 [6.27, 13.24]
P-ANMF, one bin	2.52 [0.80, 6.10]	6.59 [3.88, 11.12]
PAMF-H, one Doppler	2.18 [1.18, 3.78]	8.49 [5.83, 12.23]
IN-ARCP, residual bootstrap	—	1.22 [0.52, 1.93]

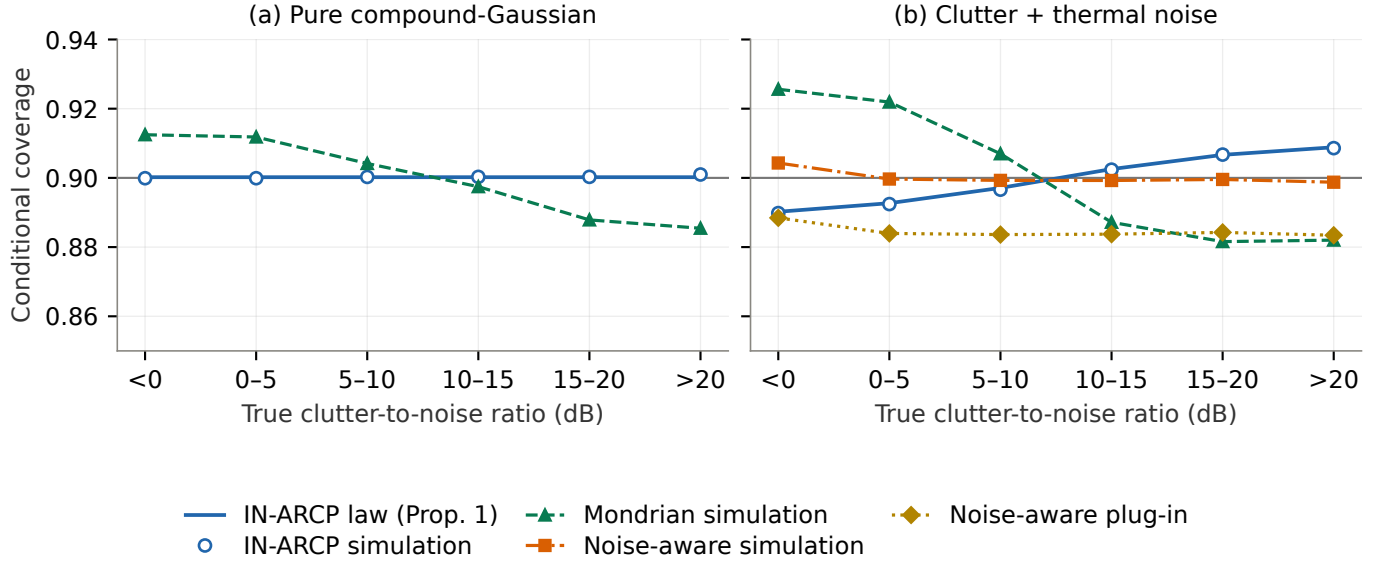


Figure S2: Synthetic verification: coverage conditional on the true clutter-to-noise ratio. The blue line is the exact law of Proposition 1; open circles are its simulation. Other markers show the comparison methods. Each panel uses 200 replications with $m = 16$ at nominal coverage 0.90; Monte Carlo standard error is at most 0.0014.

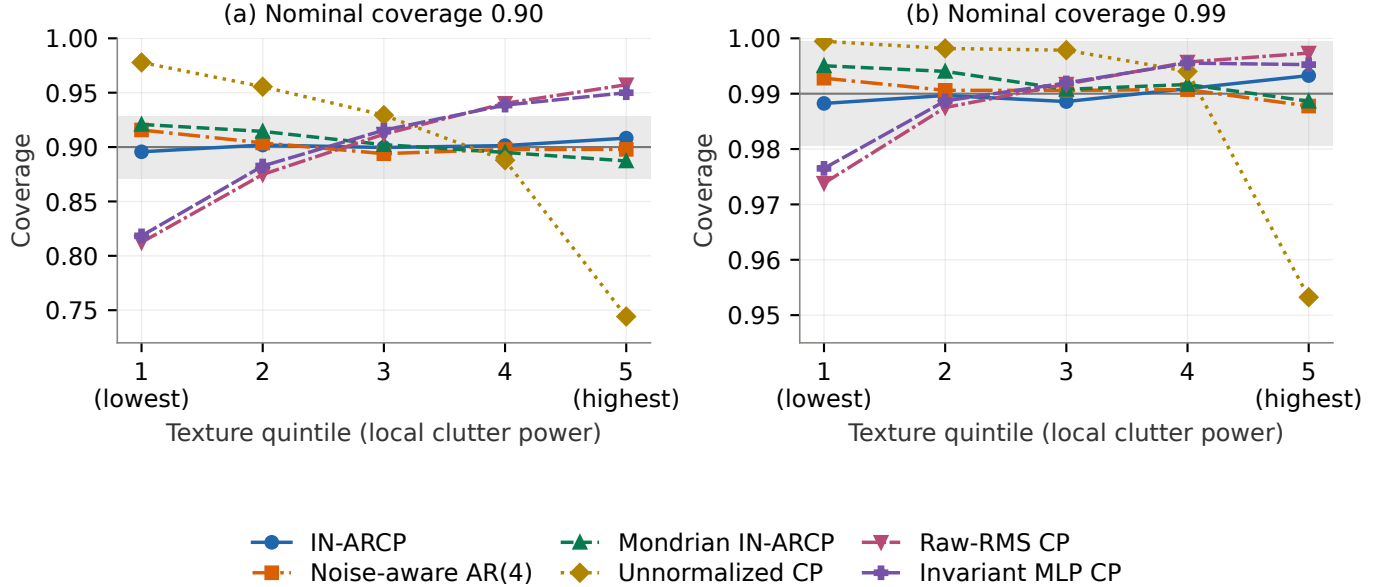


Figure S3: Coverage by texture quintile on IPIX (rotations 2-3, $m = 16$). Texture is proxied by the local clutter power (± 1024 pulses, excluding the episode). Shading shows the expected maximum quintile deviation from sampling noise alone; it is a reference band, not a confidence interval for the plotted curves.

Table S4: Texture-quintile spread of the measured false-alarm rate (max/min over quintiles of local clutter power, ± 1024 -pulse proxy), IPIX, 56 units, from `r12/analyze_r12.py` Part L. Conformal: split-conformal threshold. Model: the rule named in Table I. The clipped integrator’s conformal spread at 10^{-4} is unbounded because some quintiles record no false alarm at all, its statistic having saturated.

α	Conformal			Model-based rule		
	10^{-2}	10^{-3}	10^{-4}	10^{-2}	10^{-3}	10^{-4}
IN-ARCP (CA law)	1.3	1.5	2.7	1.3	1.4	2.0
IN-AR(4) (CA law)	1.1	1.1	1.9	1.1	1.1	1.3
OS scale, $k = 8$ (OS law)	1.7	1.9	3.0	2.1	4.0	7.1
NA-AR(4) (Gaussian)	1.9	3.0	2.9	1.6	2.3	3.2
Power, CA (white law)	1.1	1.1	1.3	1.0	1.3	1.6
Power, OS (white law)	1.1	1.9	3.3	1.1	1.4	1.8
Integration (dwell law)	1.7	1.8	2.2	1.6	2.1	2.0
PAMF-H, one Doppler (CA law)	2.2	2.1	4.0	2.2	2.4	2.4
P-ANMF, one bin (Beta)	1.5	1.7	2.0	2.3	2.4	2.0
P-ANMF, max (Fisher’s g)	4.4	9.9	23	2.3	4.5	7.0
ANMF-SCM (Kraut–Scharf)	2.3	1.8	3.2	2.2	2.0	1.8
ANMF-Tyler (FP law)	2.6	2.1	2.0	2.5	2.2	1.8
Clipped, OS scale (Gaussian sim.)	2.0	12	$> 10^4$	1.2	1.3	1.5

Table S5: IPIX width and coverage ($m = 16$; rotations 2–3; 56 units). Radius ratio: geometric mean over units of mean disc radius relative to IN-ARCP on the same test episodes, with 95% day-cluster bootstrap interval. Max. dev.: mean over units of the largest absolute texture-quintile coverage deviation from nominal. At equal model order, NA-AR(4) has 0.891 [0.872, 0.905] times the radius of the innovation-normalized AR(4) disc.

Method	$1 - \alpha = 0.90$			$1 - \alpha = 0.99$		
	Radius ratio [95% CI]	Coverage	Max. dev.	Radius ratio [95% CI]	Coverage	Max. dev.
Noise-aware AR(4) (proposed)	0.753 [0.714, 0.815]	0.902	0.043	0.743 [0.706, 0.798]	0.991	0.011
Noise-aware AR(2) (proposed)	0.777 [0.748, 0.824]	0.901	0.044	0.769 [0.745, 0.806]	0.991	0.011
AR(4), innovation-normalized	0.845 [0.805, 0.905]	0.901	0.032	0.861 [0.822, 0.918]	0.990	0.011
AR(2), innovation-normalized	0.895 [0.855, 0.953]	0.901	0.034	0.898 [0.857, 0.953]	0.990	0.010
Noise-aware AR(1)	0.966 [0.930, 0.994]	0.901	0.042	0.952 [0.907, 0.989]	0.990	0.011
IN-ARCP, AR(1)	1	0.901	0.036	1	0.990	0.011
Gaussian plug-in of IN-ARCP	0.995 [0.988, 1.002]	0.900	0.034	0.993 [0.979, 1.005]	0.990	0.011
Mondrian IN-ARCP	0.980 [0.975, 0.984]	0.904	0.043	0.998 [0.988, 1.006]	0.992	0.011
Learned-scale CP	0.921 [0.875, 0.990]	0.900	0.039	1.072 [1.019, 1.147]	0.991	0.012
Invariant MLP CP	1.337 [1.288, 1.439]	0.901	0.096	1.585 [1.489, 1.780]	0.990	0.019
Raw-RMS normalized CP	1.368 [1.320, 1.449]	0.899	0.103	1.609 [1.533, 1.782]	0.989	0.021
Unnormalized CP	1.048 [1.010, 1.085]	0.899	0.165	1.475 [1.261, 1.720]	0.989	0.039
Gaussian plug-in of noise-aware AR(4)	0.726 [0.687, 0.787]	0.885	0.049	0.698 [0.659, 0.758]	0.985	0.016
Sampling-noise reference for max. dev.			0.029			0.009

9.3 Day transfer and texture-proxy sensitivity

Texture-proxy sensitivity (pre-specified windows; mean maximum quintile deviation at $1 - \alpha = 0.90$, $m = 16$): with ± 512 pulses IN-ARCP 0.035, NA-AR(4) 0.043, unnormalized 0.181, neural center 0.103; with ± 2048 pulses IN-ARCP 0.033, unnormalized 0.141. The ranking is the same as with ± 1024 .

9.4 Hold-out series

Two further McMaster series (#269 and #287, the widely used “high” and “low” sea-state files, one provider-preprocessed VV cell each, from the same 1993 campaign) were hashed and left unopened until the main analysis was complete, and then evaluated once (Table S7). They confirm the direction of the main result, with larger gains. They also show a case in which the classical Gaussian threshold over-covers while conformal IN-ARCP stays near nominal. Their test sets are small (about 165 episodes per unit), so the hold-out is a directional check. Mondrian calibration returns infinite regions at $1 - \alpha = 0.99$ there, because each bin receives fewer calibration episodes than the rank requires.

Table S6: Day transfer ($m = 16$, $1 - \alpha = 0.90$): fit and calibrate on other days, test on each of six held-out days.

Method	Mean coverage	Worst held-out day	Radius ratio
Noise-aware AR(4) (proposed)	0.898	0.863	0.936
Noise-aware AR(2) (proposed)	0.898	0.883	0.956
AR(4), innovation-normalized	0.901	0.894	0.993
AR(2), innovation-normalized	0.899	0.891	1.024
Noise-aware AR(1)	0.900	0.882	1.037
IN-ARCP, AR(1)	0.903	0.882	1.000
Gaussian plug-in of IN-ARCP	0.906	0.893	1.006
Mondrian IN-ARCP	0.907	0.881	0.984
Learned-scale CP	0.898	0.774	0.967
Invariant MLP CP	0.880	0.622	1.200
Raw-RMS normalized CP	0.906	0.681	1.466
Unnormalized CP	0.909	0.716	1.470
Gaussian plug-in of noise-aware AR(4)	0.876	0.842	0.893

Table S7: Hold-out series #269 and #287 ($1 - \alpha = 0.90$; six units per column).

Method	$m = 8$		$m = 16$	
	Radius ratio	Coverage (min)	Radius ratio	Coverage (min)
Noise-aware AR(4) (proposed)	0.454	0.890 (0.816)	0.498	0.893 (0.847)
Noise-aware AR(2) (proposed)	0.541	0.887 (0.828)	0.533	0.895 (0.859)
AR(4), innovation-normalized	0.546	0.908 (0.847)	0.515	0.906 (0.883)
AR(2), innovation-normalized	0.582	0.905 (0.840)	0.572	0.906 (0.871)
Noise-aware AR(1)	0.921	0.898 (0.840)	0.921	0.903 (0.859)
IN-ARCP, AR(1)	1.000	0.906 (0.853)	1.000	0.910 (0.859)
Gaussian plug-in of IN-ARCP	1.178	0.949 (0.896)	1.104	0.945 (0.895)
Mondrian IN-ARCP	1.000	0.895 (0.804)	0.981	0.896 (0.810)
Learned-scale CP	0.629	0.893 (0.804)	0.662	0.919 (0.791)
Invariant MLP CP	1.809	0.884 (0.736)	2.158	0.903 (0.761)
Raw-RMS normalized CP	1.321	0.880 (0.706)	1.243	0.887 (0.706)
Unnormalized CP	1.142	0.868 (0.607)	1.161	0.859 (0.589)
Gaussian plug-in of noise-aware AR(4)	0.411	0.866 (0.816)	0.472	0.856 (0.755)

9.5 Dependence, day sensitivity and optimizer diagnostics

At $\alpha = 0.1$, mean lag-one exceedance autocorrelation within an IPIX bin (256-pulse separation) is -0.007 , with unit range $[-0.067, 0.046]$; same-time cross-bin correlation is 0.045 . These diagnostics do not prove independence. Observed maximum quintile coverage deviations for IN-ARCP, IN-AR(4), NA-AR(4) and the unnormalized score are 0.036 , 0.032 , 0.043 and 0.165 . The binomial reference is 0.029 ; dependence-aware time-shift references are 0.033 , 0.033 , 0.039 and 0.062 .

The AR(1) clamp binds in 5 of 56 IPIX units. NA-AR(2) converges in every unit; NA-AR(4) converges in 16 at the 1,500-iteration cap and 55 after a further 4,500 iterations. The exploratory restart changes its radius by geometric factors 0.9989 at coverage 0.90 (unit range 0.9790 – 1.0132) and 0.9997 at 0.99 (0.9492 – 1.0327); coverage changes from 0.9019 to 0.9020 and from 0.9905 to 0.9906 . Median likelihood gain per training episode is 3.13×10^{-6} , maximum 1.54×10^{-4} . The median whole-unit GPU time for thirteen procedures is 71 s, not a detector hardware benchmark.

The six daily onset gains of IN-ARCP over local power at $P_d = 0.5$ are 3.5, 8.4, 13.8, 13.3, 12.2 and 11.4 dB. They should not be read as interchangeable independent replications.

9.6 Parametric texture and adaptive conformal baselines

A fitted K-texture law changes NA-AR(4)’s radius by factors 1.0092 [1.0064 , 1.0115] at coverage 0.90 and 1.0078 [1.0034 , 1.0104] at 0.99 ; coverage and quintile deviations are essentially unchanged. Within-session ACI with $\gamma = 0.005$ or 0.02 gives similar coverage and finite radii; at 0.99 , $\gamma = 0.02$ gives infinite regions in 0.0115 of episodes. Day-transfer worst coverages improve from 0.8822 for split IN-ARCP to 0.8889 and 0.8965 at coverage 0.90 , and from 0.9875 to 0.9889 and 0.9902 at 0.99 . Clean-feedback behavior does not establish safety under target-contaminated online feedback: at 5% contamination ungated ACI gives 0.14α with $P_d = 0.275$ (split: 0.890); gated ACI still gives only 0.55α or 0.68α . Full settings, trajectories and uncertainty remain in the baseline and R10 outputs.

Table S8: Leave-one-day-out ranges from `r10/audit_existing.txt`. Gains are dB; radius ratio and spread are dimensionless. These complement the approximate six-day cluster intervals.

Effect	All six days	LODO range
NA-AR(4)/IN-AR(4) radius ratio (0.90)	0.891	0.884–0.895
IN1 quintile Pfa spread ($\alpha = 0.01$)	$1.75\times$	1.72–1.83x
single-pulse gain IN1 vs CA16 $P_d = 0.5$	6.96 dB	5.97– 8.16
single-pulse gain IN1 vs CAlloc $P_d = 0.5$	9.81 dB	8.94–11.54
single-pulse gain NA4 vs IN1 $P_d = 0.5$	1.07 dB	1.03– 1.11
single-pulse gain NA4 vs IN4 $P_d = 0.5$	0.66 dB	0.58– 0.75
single-pulse gain IN1 vs CA16 $P_d = 0.9$	7.62 dB	6.59– 9.01
single-pulse gain IN1 vs CAlloc $P_d = 0.9$	8.97 dB	8.10–10.62
single-pulse gain NA4 vs IN1 $P_d = 0.9$	2.19 dB	1.88– 2.39
single-pulse gain NA4 vs IN4 $P_d = 0.9$	1.46 dB	1.43– 1.52
eight-look dwell gain IN1 vs CAlloc-sum $P_d = 0.5$	10.70 dB	9.86–12.38
eight-look dwell gain NA4 vs CAlloc-sum $P_d = 0.5$	11.74 dB	11.01–13.00

10 Detection comparisons and target-history contamination

Injected comparisons share known timing and target-aligned dwells. They test mechanisms, not operational acquisition. Figure S4 retains the 10^{-3} design rate and full method curves.

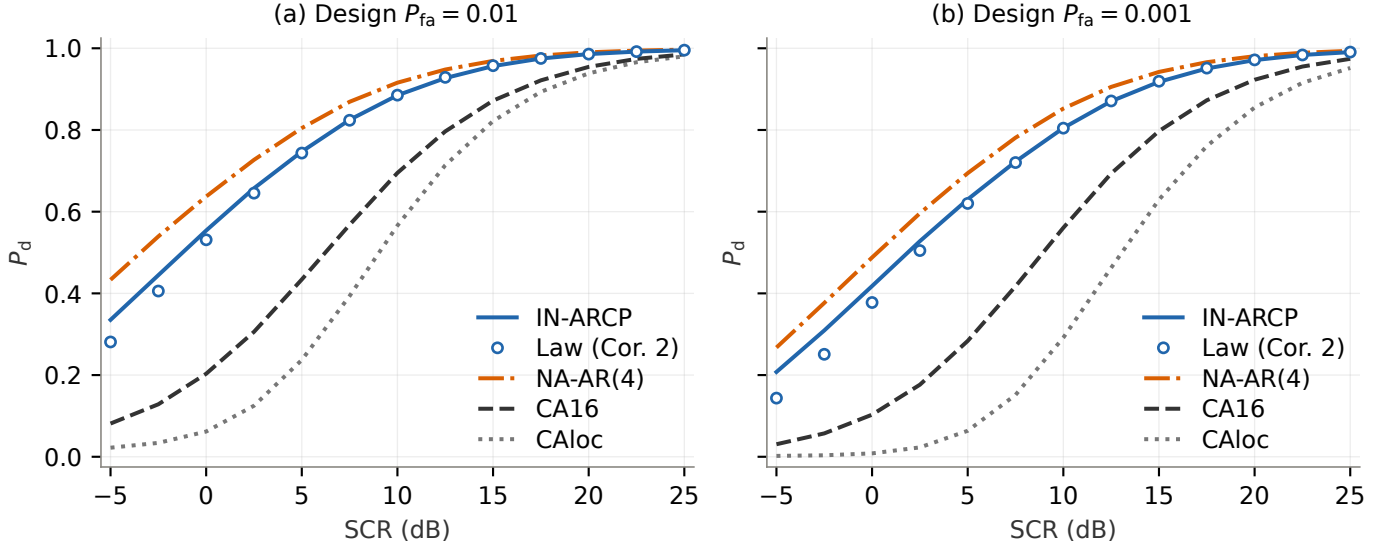


Figure S4: Injected Swerling-1 onset detection on IPIX at design false-alarm probabilities 0.01 and 0.001 ($m = 16$, conformal thresholds). Empirical and law curves are unweighted unit means; circles show the fitted-model law of Corollary 2. Reported SCR gains average per-unit gains.

11 Dwell detection on IPIX

PAMF-H has the best tested dwell sensitivity. Its endpoint at the lowest tested SCR makes the ANMF-Tyler gap of 5.36 [1.93, 8.83] dB a lower bound. Clipping costs 2.04 [0.00, 2.90] dB against AR(1) integration; the onset raw-power-OS/IN-ARCP gap is 10.62 [7.05, 12.69] dB. The earlier R10 equal-dwell comparison gives PAMF-H a 1.0 [0.3, 1.7] dB advantage over integrated IN-ARCP at $P_d = 0.5$ (about 4 dB at 0.9); the studies have separate calibrations. A Hann-windowed range-CA Doppler bank needs 5.0 [4.2, 6.0] dB more SCR than integrated IN-ARCP. PAMF-H's matched-Doppler cost in that study is 4.25 dB.

Table S9: Onset SCR gain at design $P_{fa} = 0.01$: mean per-unit gain [95% day-cluster interval], from `detection/detection.summary.json`. Positive means less SCR. Local power uses the non-causal ± 1024 -pulse window.

Method	P_d	Gain over CA16 (dB)	Gain over local power (dB)
IN1	0.5	6.96 [4.29, 9.04]	9.81 [6.41, 12.61]
IN4	0.5	7.37 [4.54, 9.52]	10.22 [6.67, 13.26]
NA4	0.5	8.03 [5.45, 10.02]	10.88 [7.56, 13.66]
G1	0.5	7.05 [4.40, 9.11]	9.90 [6.48, 12.71]
U	0.5	2.30 [0.78, 3.44]	5.15 [3.34, 6.61]
MLP	0.5	3.88 [1.89, 5.34]	6.73 [3.89, 8.70]
LS	0.5	5.72 [2.75, 7.96]	8.57 [4.89, 11.49]
IN1	0.9	7.62 [4.55, 10.01]	8.97 [5.58, 11.67]
IN4	0.9	8.35 [4.95, 10.96]	9.70 [5.90, 12.64]
NA4	0.9	9.81 [6.27, 12.45]	11.16 [7.28, 14.14]
G1	0.9	7.67 [4.71, 10.02]	9.02 [5.70, 11.73]
U	0.9	2.80 [1.01, 4.34]	4.15 [2.23, 5.73]
MLP	0.9	4.36 [2.30, 5.85]	5.71 [3.23, 7.35]
LS	0.9	6.14 [2.75, 8.69]	7.49 [3.78, 10.46]

Table S10: SCR (dB) for $P_d = 0.5$ and 0.9 over an 8-pulse dwell: persistent Swerling-1 target with random Doppler from the first dwell pulse (IPIX, conformal thresholds). ≤ -5 : reached at the lowest SCR tested; n.r.: not reached at 25 dB.

Detector	$\alpha = 0.01$		0.001
	$P_d = 0.5$	0.9	0.5
PAMF-H	≤ -5	5.1	-4.3
Integration, AR(4)	≤ -5	7.0	-1.6
Integration, AR(1)	-4.4	8.3	-0.1
Clipped, OS scale	-2.3	13.0	n.r.
Binary, OS scale	-1.2	14.5	n.r.
ANMF-SCM	-0.2	20.3	9.5
ANMF-Tyler	0.4	20.3	9.8
P-ANMF	0.2	n.r.	n.r.
IN-ARCP, first look	-1.2	11.0	1.9
Power OS, first look	9.4	19.5	14.3

12 Guarded scale, clip level, certified trimming, pulse blanking and bursts (R14)

Frozen protocol `r14/PROTOCOL_R14.md` (sha256 9f771091...), written after a synthetic dry run and before any IPIX outcome; expectations and verdicts in `r14/RESULTS_R14.md` (12 of 15 met). The settings are those of the final IPIX study, except for dense calibration: non-overlapping 24-pulse segments, about 14,600 per unit, so that $\alpha = 10^{-3}$ is resolved. The study tested five changes:

- a guard of $\Delta \in \{0, 8, 16\}$ pulses between the scale window and the tested pulse;
- clip levels $\kappa \in \{4, 6, 9, 12, 18\}$;
- a certified trimmed sum, which discards the largest dwell terms; the number discarded is the smallest for which more corrupted dwell innovations than discarded terms have probability at most $\alpha/5$ under the hit model (4 of eight at a 2% and 5 at a 5% hit rate at $\alpha = 10^{-2}$; no such number exists for bursts);
- pulse blanking of isolated spikes, calibrated with masks drawn from the hit model and passed through the same rule (mask-matched calibration);
- bursts of eight consecutive pulses at the same mean hit rate.

The certified thresholds use the worst case of Theorem 1 with hits drawn either from the independent-hit model or from the burst model. The guard leaves the false-alarm law unchanged; the measured P_{fa}/α with $\Delta = 8$ is 1.03 at 10^{-2} and 1.03 at 10^{-3} . The real IPIX target stays undetected with every guard (exceedance 0.0005 at 10^{-3} with $\Delta = 8$), because it is present in every history.

Unmet expectations of R14.

- **Onset cost (G3).** The guard was expected to cost at most 0.5 dB at onset. It costs 0.9 dB with $\Delta = 8$ and 1.2 dB with $\Delta = 16$, because the scale is older.
- **Certification at 10^{-3} (K2).** The certificate at the clip level chosen by the saturation rule was expected to reach $P_d = 0.5$ below 25 dB in all four interference conditions at 10^{-3} . With 5% hits the certificate is vacuous at 10^{-3} for every clip level.
- **Cost of trimming (T2).** Trimming was expected to cost more than clipping at both hit rates. At 5% it costs +5.1 dB against +5.9 dB for clipping ($\kappa = 6$).

The SCR for $P_d = 0.5$ is fragile under 5% hits, because the certified detection curve is nearly flat there, so the main text reports P_d at 10 dB.

Table S11: Guarded scale: clean rate ratio [cluster interval], and mean per-look P_d over looks 1–8 at 10 dB. R14 IPIX and R15 NetRAD; these eight looks do not measure post-guard persistence.

Radar	Δ	$\alpha = 10^{-2}$	10^{-3}	Random	Opposite	Matched
IPIX	0	1.01 [0.98, 1.04]	1.02 [0.94, 1.10]	0.24	0.36	0.01
IPIX	8	1.03 [0.97, 1.07]	1.03 [0.84, 1.21]	0.82	0.95	0.07
IPIX	16	1.00 [0.93, 1.03]	1.03 [0.81, 1.15]	0.82	0.94	0.07
NetRAD	0	1.05 [0.98, 1.14]	1.22 [0.94, 1.65]	0.30	0.43	0.00
NetRAD	8	1.05 [1.00, 1.11]	1.23 [0.98, 1.64]	0.92	0.99	0.03
NetRAD	16	1.05 [1.00, 1.12]	1.36 [1.05, 1.87]	0.92	0.99	0.03

Table S12: Detection within eight looks for linear emergence, 10 dB SCR and design $P_{fa} = 0.01$.

Radar	Ramp length	$\Delta = 0$	8	16
IPIX	8	0.745	0.908	0.909
IPIX	16	0.032	0.869	0.903
NetRAD	8	0.862	0.964	0.966
NetRAD	16	0.016	0.937	0.959

The earlier pre-onset ramp comparison, corrected R10 verdict, and all R11 failed expectations remain in Section 3. Exact ramp-law error is 0.016, within the 0.03 expectation. The exploratory post-onset law has mean per-unit/look errors 0.010, 0.003 and 0.008 for random, matched and opposite Doppler. Cumulative-null and equal-dwell comparisons remain in the full onset outputs. The R11 median OS score has onset cost 1.99 [0.89, 3.39] dB, quintile spread 2.06 and coverage-law error 0.0214; these miss the stated tolerances.

Table S13: Clean clipped integration, IPIX R14 dense calibration: rate ratio [95% day-cluster interval].

κ	$\alpha = 10^{-2}$	10^{-3}
4	1.04 [0.97, 1.08]	0.09 [0.00, 0.23]
6	1.05 [0.99, 1.08]	0.67 [0.35, 0.94]
9	1.05 [0.98, 1.09]	0.90 [0.58, 1.15]
12	1.06 [1.00, 1.09]	0.95 [0.65, 1.20]
18	1.06 [1.01, 1.09]	1.03 [0.79, 1.19]

The pre-specified saturation rule chooses $\kappa = 9$ in 17 of 56 units at 10^{-3} and finds no eligible tested level ($\kappa \leq 18$) in 16. Applying $\kappa = 9$ to every unit is post hoc: at 10^{-3} it changes 10-dB detection from 0.46 to 0.68. Sparse R12 and dense R14 calibrations are separate analyses; their endpoints must not be combined.

At 10^{-3} and 5% independent hits, every tested clip certificate is vacuous. At 10^{-2} and 5% hits, trimming costs 5.1 dB versus clipping's 5.9 dB for $\kappa = 6$, contrary to the expectation that trimming costs more. Fixed-SCR detection is clearer because the certified curves are nearly flat. Burst-matched trimming has no usable choice. The R12 sparse-calibration binary certificate has 10-dB detection within 0.03 of certified clipping at both tested rates and powers. At design $\alpha = 10^{-2}$, uncertified integration and PAMF-H exceed design 10–23-fold, while self-normalized statistics stay within 1.21 times design at a sensitivity cost. Full clip grids, SCR costs and sparse-calibration rows remain in the frozen outputs.

Table S14: IPIX independent-hit remedies at $\alpha = 10^{-2}$: rate ratio [95% day-cluster interval], detection at 10 and 25 dB SCR. R14 dense calibration; interference is 10 or 30 dB above local power. Certificate calibration uses matching independent hits.

Hit rate	JNR (dB)	Statistic	Rate ratio	$P_d(10)$	$P_d(25)$
2%	10	Clean clip6	1.08 [1.05, 1.11]	0.85	0.99
2%	10	Certified clip6	0.53 [0.43, 0.61]	0.81	0.95
2%	10	Certified clip9	0.52 [0.44, 0.58]	0.81	0.96
2%	10	Certified trim4	0.53 [0.48, 0.57]	0.79	0.97
2%	10	Blanking	1.05 [1.01, 1.08]	0.79	0.97
2%	30	Clean clip6	1.10 [1.07, 1.12]	0.85	0.99
2%	30	Certified clip6	0.55 [0.46, 0.61]	0.81	0.95
2%	30	Certified clip9	0.53 [0.45, 0.58]	0.81	0.96
2%	30	Certified trim4	0.57 [0.51, 0.60]	0.79	0.97
2%	30	Blanking	1.08 [1.03, 1.11]	0.79	0.97
5%	10	Clean clip6	1.22 [1.15, 1.31]	0.85	0.98
5%	10	Certified clip6	0.12 [0.09, 0.14]	0.64	0.75
5%	10	Certified clip9	0.13 [0.08, 0.17]	0.71	0.87
5%	10	Certified trim5	0.15 [0.12, 0.17]	0.72	0.95
5%	10	Blanking	1.13 [1.04, 1.21]	0.78	0.96
5%	30	Clean clip6	1.28 [1.21, 1.37]	0.85	0.98
5%	30	Certified clip6	0.12 [0.10, 0.14]	0.64	0.75
5%	30	Certified clip9	0.14 [0.10, 0.18]	0.71	0.87
5%	30	Certified trim5	0.23 [0.19, 0.26]	0.72	0.95
5%	30	Blanking	1.21 [1.16, 1.25]	0.78	0.96

Table S15: IPIX independent-hit remedies at $\alpha = 10^{-3}$: rate ratio [95% day-cluster interval], detection at 10 and 25 dB SCR. R14 dense calibration; interference is 10 or 30 dB above local power. Certificate calibration uses matching independent hits.

Hit rate	JNR (dB)	Statistic	Rate ratio	$P_d(10)$	$P_d(25)$
2%	10	Clean clip6	0.65 [0.34, 0.92]	0.46	0.54
2%	10	Certified clip6	0.07 [0.03, 0.11]	0.12	0.16
2%	10	Certified clip9	0.27 [0.17, 0.34]	0.52	0.63
2%	10	Certified trim5	0.39 [0.30, 0.44]	0.68	0.94
2%	10	Blanking	1.09 [1.00, 1.17]	0.68	0.93
2%	30	Clean clip6	0.65 [0.33, 0.90]	0.46	0.54
2%	30	Certified clip6	0.07 [0.03, 0.11]	0.12	0.16
2%	30	Certified clip9	0.26 [0.16, 0.33]	0.52	0.63
2%	30	Certified trim5	0.43 [0.33, 0.49]	0.68	0.94
2%	30	Blanking	1.13 [1.04, 1.18]	0.68	0.93
5%	10	Clean clip6	0.66 [0.35, 0.94]	0.45	0.54
5%	10	Certified clip6	0.00 [0.00, 0.00]	0.00	0.00
5%	10	Certified clip9	0.00 [0.00, 0.00]	0.00	0.00
5%	10	Certified trim6	0.00 [0.00, 0.00]	0.00	0.00
5%	10	Blanking	1.67 [1.39, 1.91]	0.67	0.93
5%	30	Clean clip6	0.69 [0.39, 0.93]	0.45	0.54
5%	30	Certified clip6	0.00 [0.00, 0.00]	0.00	0.00
5%	30	Certified clip9	0.00 [0.00, 0.00]	0.00	0.00
5%	30	Certified trim6	0.00 [0.00, 0.00]	0.00	0.00
5%	30	Blanking	2.25 [1.87, 2.54]	0.67	0.93

Table S16: Eight-pulse bursts on IPIX at JNR 30 dB: both hit rates and both design rates, including burst-matched calibration. Burst certificates also detect no target at 25 dB.

Hit rate	α	Statistic	Rate ratio [95% CI]	$P_d(10)$
2%	10^{-2}	Clean clip6	2.24 [2.11, 2.33]	0.84
2%	10^{-2}	Independent-hit cert., clip6	1.46 [1.33, 1.58]	0.80
2%	10^{-2}	Independent-hit cert., clip9	1.75 [1.59, 1.89]	0.80
2%	10^{-2}	Burst certificate, clip6	0.00 [0.00, 0.00]	0.00
2%	10^{-2}	Blanking, independent masks	2.31 [2.22, 2.36]	0.79
2%	10^{-2}	Blanking, burst masks	2.25 [2.17, 2.31]	0.79
2%	10^{-3}	Clean clip6	3.86 [2.11, 5.34]	0.45
2%	10^{-3}	Independent-hit cert., clip6	0.90 [0.38, 1.61]	0.12
2%	10^{-3}	Independent-hit cert., clip9	3.86 [2.78, 4.99]	0.51
2%	10^{-3}	Burst certificate, clip6	0.00 [0.00, 0.00]	0.00
2%	10^{-3}	Blanking, independent masks	13.60 [12.63, 14.17]	0.69
2%	10^{-3}	Blanking, burst masks	13.43 [12.42, 14.00]	0.68
5%	10^{-2}	Clean clip6	3.98 [3.65, 4.22]	0.82
5%	10^{-2}	Independent-hit cert., clip6	1.25 [1.04, 1.45]	0.62
5%	10^{-2}	Independent-hit cert., clip9	1.66 [1.44, 1.83]	0.69
5%	10^{-2}	Burst certificate, clip6	0.00 [0.00, 0.00]	0.00
5%	10^{-2}	Blanking, independent masks	4.14 [4.01, 4.23]	0.78
5%	10^{-2}	Blanking, burst masks	4.10 [3.98, 4.17]	0.78
5%	10^{-3}	Clean clip6	8.92 [5.10, 12.32]	0.44
5%	10^{-3}	Independent-hit cert., clip6	0.00 [0.00, 0.00]	0.00
5%	10^{-3}	Independent-hit cert., clip9	0.00 [0.00, 0.00]	0.00
5%	10^{-3}	Burst certificate, clip6	0.00 [0.00, 0.00]	0.00
5%	10^{-3}	Blanking, independent masks	32.43 [30.34, 33.64]	0.68
5%	10^{-3}	Blanking, burst masks	32.35 [30.32, 33.49]	0.67

12.1 Pulse blanking: the thresholds

Pulse blanking discards an innovation power, and the innovation after it, when the power exceeds both ten times the history median and four times a robust level of its block. That level is the history median for history terms and the third-smallest dwell power for dwell terms. The remaining dwell powers are integrated. Blanking is calibrated on clean episodes blanked at hit positions drawn from the hit model, so it has a conformal threshold but no certificate: nothing bounds the statistic when a hit survives the blanking rule.

13 77 GHz thermal-noise and real-interference checks

Range bins 8–247 and receivers 0, 5, 10 and 15 of both stations are used. Figure S5 shows coverage and onset gain by CNR class; the general-form gain curve is post hoc. IN-ARCP’s gain over local power rises from -1.45 dB in noise-dominated cells to 3.11 dB at 10 – 20 dB CNR. Only 338 test episodes from two units occupy the highest CNR class, versus 721,095 in the lowest. At design 0.01 its CNR-class rate ratios are 1.07, 1.06, 0.95, 0.52, 0.08 and 0 across < -5 , $[-5, 0)$, $[0, 5)$, $[5, 10)$, $[10, 20)$ and ≥ 20 dB; the support is uneven.

For real interference, calibration masks are random 24-chirp blocks from frames 0–49; tests use frames 50–99. N: no fast-time mitigation; Z: zeroing; ZAR: zeroing with AR(8) reconstruction. Test hit rates are 45.4% in dense A and 4.6% in sparse C. The empirical masks do not establish exchangeability, clutter independence or inclusion dominance. In sparse C without mitigation, integration runs at 3.56 times design in the 1–2-hit stratum. The clipped certificate costs 3.1 dB on hit-free segments and 6.4 dB overall against clean-threshold clipping. In dense A it never reaches $P_d = 0.5$ overall. AR reconstruction improves IN-ARCP SCR50 by only 0.18 dB, below the expected 1 dB.

14 The real IPIX target: detection versus dwell length

Calibration of the self-normalized detector degrades for longer, overlapping windows. Its analytic Fisher’s g threshold fails already at $N = 8$, consistent with imperfect whitening by an AR(4) model fitted on neighboring cells. The value of P-ANMF here is a calibrated short-dwell threshold, not sensitivity.

History-normalized IN-ARCP and clipping exceed on the primary target cell at only 0.0004 at design 10^{-3} . This supports full-history self-masking, not real-onset validation. The long-dwell P-ANMF rate exceeds twice design; no i.i.d.

Table S17: Real 77 GHz interference, $\alpha = 0.01$: all three mitigations. The last column is the hit-free stratum, not a confidence interval; n.r.: not reached in the tested SCR range. Full per-hit-stratum counts and ratios remain in `r12/r12_jku_summary.txt`.

Run	Mitigation	Statistic	Rate ratio	SCR50 all	SCR50 clean
ref	N	Integration	1.03	5.91	5.91
ref	N	Clean clip	1.00	6.84	6.84
ref	N	Certified clip	1.00	6.84	6.84
ref	N	Certified binary	0.67	7.96	7.96
ref	Z	Integration	0.99	6.24	6.24
ref	Z	Clean clip	0.97	7.11	7.11
ref	Z	Certified clip	0.97	7.11	7.11
ref	Z	Certified binary	0.63	8.32	8.32
ref	ZAR	Integration	1.01	6.09	6.09
ref	ZAR	Clean clip	0.99	7.00	7.00
ref	ZAR	Certified clip	0.99	7.00	7.00
ref	ZAR	Certified binary	0.66	8.17	8.17
A	N	Integration	0.81	11.14	7.68
A	N	Clean clip	1.01	11.95	8.66
A	N	Certified clip	0.11	n.r.	13.74
A	N	Certified binary	0.06	n.r.	14.93
A	Z	Integration	0.96	6.96	6.11
A	Z	Clean clip	0.98	7.86	7.05
A	Z	Certified clip	0.11	n.r.	12.24
A	Z	Certified binary	0.06	n.r.	13.63
A	ZAR	Integration	1.01	6.74	6.06
A	ZAR	Clean clip	1.01	7.65	7.03
A	ZAR	Certified clip	0.11	n.r.	12.18
A	ZAR	Certified binary	0.06	n.r.	13.52
C	N	Integration	1.42	7.15	6.34
C	N	Clean clip	1.29	8.11	7.29
C	N	Certified clip	0.54	14.52	10.40
C	N	Certified binary	0.33	15.78	11.63
C	Z	Integration	1.00	6.41	6.22
C	Z	Clean clip	1.02	7.26	7.08
C	Z	Certified clip	0.51	13.94	10.00
C	Z	Certified binary	0.27	15.19	11.33
C	ZAR	Integration	1.04	6.25	6.10
C	ZAR	Clean clip	1.02	7.15	7.01
C	ZAR	Certified clip	0.50	13.88	9.93
C	ZAR	Certified binary	0.30	15.08	11.22

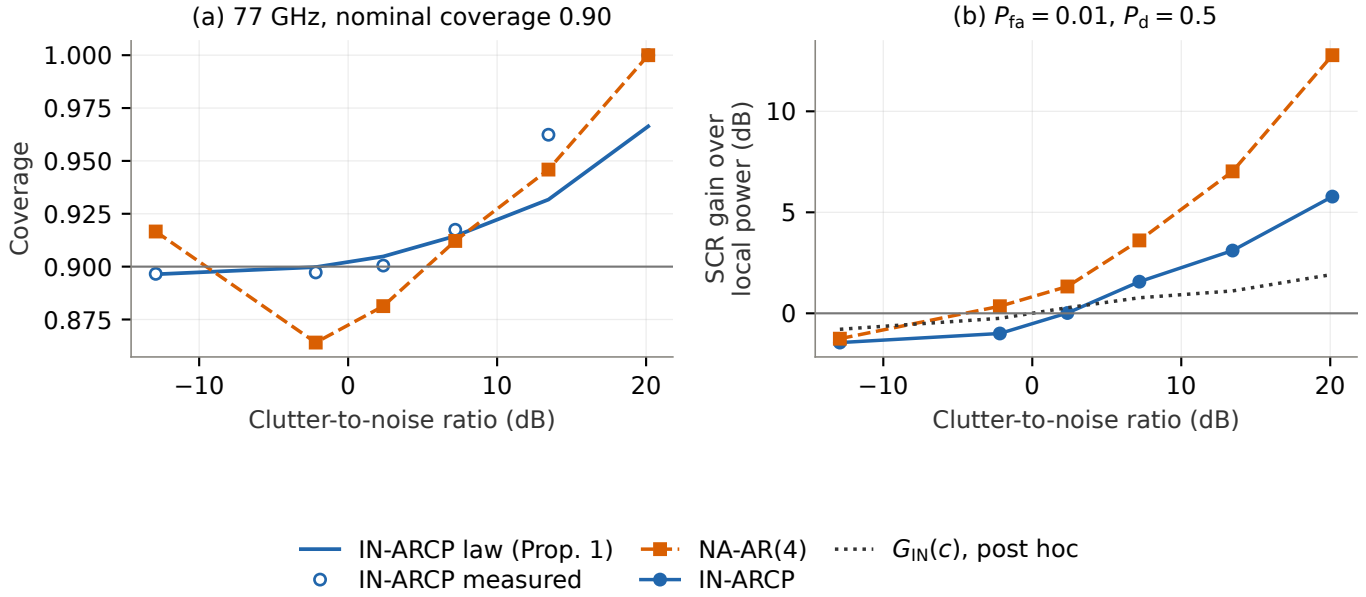


Figure S5: 77 GHz ground clutter: (a) coverage by CNR class at nominal 0.90; (b) onset SCR gain over local power for $P_d = 0.5$ at design $P_{fa} = 0.01$. The general-form gain interpretation is post hoc. CNR classes have unequal support, detailed in the text.

Table S18: Real IPIX target (primary cell, test thirds, 56 units): mean P_d at design $P_{fa} = 10^{-3}$ versus dwell N . In parentheses: measured clutter $P_{fa} (\times 10^{-3})$. P-ANMF with Fisher's g uses the analytic threshold.

N	P-ANMF		Range CA integration	ANMF Tyler
	conformal	Fisher's g		
8	0.22 (1.1)	0.37 (13)	0.003	0.13
16	0.22 (1.3)	0.45 (32)	0.003	0.18
32	0.22 (1.3)	0.56 (69)	0.004	—
64	0.30 (1.3)	0.67 (121)	0.004	—
128	0.37 (1.2)	0.75 (178)	0.004	—
256	0.40 (1.5)	0.79 (235)	0.004	—
512	0.37 (2.0)	0.83 (299)	0.005	—
1024	0.41 (2.6)	0.87 (404)	0.007	—

Beta guarantee follows for the overlapping real-data windows. The one-second comparison with target-trained literature is descriptive because training and evaluation protocols differ.

15 Second sea-clutter radar: NetRAD (R15)

Frozen protocol `r15/PROTOCOL_R15.md` (sha256 945c5e51...). It was written before any statistic of the NetRAD samples was computed. Before that point only the following had been read: file sizes, array shapes, and the providers' scripts and log. In addition, two of the providers' own quick-look images were viewed while the dataset was being located. Expectations and verdicts are in `r15/RESULTS_R15.md` (12 of 15 met), and deviations in `r15/DEVIATIONS_R15.md`. The deviations are a corrected site description (the Atlantic side of the Cape Peninsula, not False Bay) and run mechanics; none changes a method, setting or expectation.

Data. Node 3 (monostatic) of the NetRAD system: seven HH and seven VV recordings of 130,000 pulses at 1 kHz. The clutter cells are the providers' per-recording range windows (`NR_Range_Selection_June9.m`). Each cell has its complex mean removed and is scaled to unit power. The cross-polarized recordings were set aside before any data were read. All methods are the frozen R12 and R14 code. The only change is that the ANMF secondary cells are restricted to those two to four cells away, which gives 15–30 vectors (20–35 on IPIX). The AR(1) clamp $|r| \leq 0.98$ binds in 12 of the 14 recordings.

Unmet expectations of R15.

- **Calibration at 10^{-4} (N1).** At least 12 of 13 conformal ratios were expected within $[0.5, 2]$ at 10^{-4} ; 7 were. Outside: Clip-OS1 (0.39), D-IN4 (4.26), NA4 (2.08), P-ANMF (2.33), P-ANMF(bin) (2.52), PAMF-H(w) (2.18). At 10^{-3} all 13 are within $[0.5, 2]$, and the 12 non-saturating statistics lie within 1.01–1.36 times design.
- **Pivotality ranking (N4).** IN-ARCP’s quintile spread at 10^{-2} was expected to be at most 2 and below CA16’s. It is 2.40 against 1.51. In the same analysis on IPIX it is 1.31 against 1.06, so this expectation had no support on IPIX either.
- **Flagged pulses (W2).** Non-coherent integration with a clean threshold was expected to exceed 1.5α in segments with flagged pulses. It stays at 1.08 times design.

Expectation N5 (PAMF-H within 0.5 dB of the best dwell detector) was met, but it is uninformative: every whitened dwell detector reaches $P_d = 0.5$ at the lowest SCR tested.

Exploratory (post hoc, after the outcomes and the claim audit). Two scripts look at where the conformal exceedances at 10^{-4} fall: `r15/explore_r15_hotspots.py` and `r15/explore_r15_attribution.py`.

- **Integration and PAMF-H.** Most of their excess comes from two VV units (1450 and 1501, assignment 3). Their exceedances fall within one to three seconds of the test third and coincide with flagged pulses, while their calibration thirds contain no flagged pulse. Without these two units the pooled ratios are 1.56 (integration) and 1.33 (PAMF-H).
- **P-ANMF statistics.** Their excess comes from one HH unit (1128, assignment 3) with frequent flagged pulses in both thirds. Fewer than half of its exceedances coincide with flagged pulses.
- **Unexplained.** The flagged pulses do not explain the excess of the noise-aware score (2.08). Nor do they explain an excess of integration of more than five times design in one unit that has no flagged pulse at all (1508, assignment 2).

What the flags are is not established. `r15/explore_r15_flag_extent.py` counts, in the raw recordings, the range bins above ten times their median:

- a flagged HH pulse has a median of about 108 such bins, of 1024, and 85–91% of them lie outside the clutter window;
- unflagged HH pulses with an elevated window bin have about 80.

The flags are therefore the upper tail of a phenomenon that extends over range. They may be wideband interference: the providers remove WiFi pulses in their own processing of the bistatic nodes. They may equally be range-extended HH sea spikes. The main text therefore calls them flagged pulses, not interference.

In the clutter cells, uncertified non-coherent integration runs at 1.05 times design at 10^{-2} and 1.20 at 10^{-3} over all test segments of the 7 HH recordings, the only ones with at least 0.5% flagged pulses (VV: at most 0.1%). Segments without a flagged pulse give 1.04 and 1.17; segments with flagged pulses give at most 1.08 and 1.33, with wide intervals at 10^{-3} (`r15/r15_summary.txt`, Part W). A certificate whose hit model must cover these frequent, bursty masks is nearly vacuous ($P_d = 0.04$). Blanking with mask-matched calibration keeps $P_d = 0.92$ at 0.57 times design.

The full pooled low-rate intervals remain in Table S3; removing hotspot units is not used to replace the headline results. At 10^{-3} the flagged-pulse certificate is wholly vacuous, whereas mask-matched blanking has rate ratio 0.24 (upper recording-cluster bound 0.49) and $P_d = 0.83$. Flag origin, unexplained excesses and the weak burst remedy remain unresolved.

16 Real onsets of a walking person at 77 GHz (R17)

Frozen protocol `r17/PROTOCOL_R17.md`, committed and pushed before the data were obtained; verdicts in `r17/RESULTS_R17.md`. In the pedestrian run of the 77 GHz data (`meas_3_int_A_pedestrian`, interference scenario A) a person walks in front of Station 1. Frames are 200 ms apart, so the person moves through the 15-cm range bins between frames, and each arrival of its return in a bin is a real onset in that bin’s frame-to-frame sequence. Because the receivers are real-valued, Station 1 sees the person twice: directly, and bistatically via Station 2’s transmission, about 130 bins (19 m) farther in this run. The 74 onsets are those of both returns (38 direct, 36 bistatic) in adjacent bins over 31 frames, so they are not independent arrivals.

Ground truth. Station 1’s range-Doppler map over all 128 chirps and 16 receivers (fast-time zeroing). A bin is moving in a frame when its largest power over Doppler bins at least two from zero exceeds its median over the 100 frames by 10 dB. An onset is a bin that becomes moving after 16 frames without moving energy; it is the person’s when one of the frame’s two strongest moving peaks (13 dB) lies within three bins in the onset frame or the next two.

Detectors. One chirp per frame (chirps 0, 32, 64, 96; receivers 0, 5, 10, 15: 16 looks), history of 16 frames: IN-ARCP (AR(1) per bin, fitted on clean frame pairs), the power clutter map (CA-CM: power over the mean history power), its OS form (OS-CM: eighth smallest of 16) and range CA-CFAR on the current frame (2 guard and 8 reference bins on each

side). Conformal thresholds from null episodes of even bins, false alarms on odd bins. The primary endpoint is detection within three frames of the onset.

Outcome. The protocol required onsets in at least eight distinct five-frame blocks; the person was tracked for about six seconds and the onsets fall in seven, so expectation E1 is not met and every result is descriptive. Descriptively, all lie on the side the protocol expected: false-alarm rates at design, $\text{IN-ARCP} \geq \text{CA-CM} > \text{OS-CM} > \text{range CA-CFAR}$, and IN-ARCP’s edge over CA-CM confined to the half of the onsets in bins with stronger static returns, as Proposition 2 predicts for nearly uncorrelated frame-to-frame clutter. *Limitations* (stated as the protocol requires): one walk, tracked for about six seconds, by one person; ground clutter at 77 GHz, not sea clutter; frame-rate rather than pulse-rate slow time; ground truth from the same radar’s range-Doppler processing, with no external sensor. Before the expectations were written, a mechanics check ran the whole chain on the interference run without the pedestrian, with a synthetic walker; its output, including the four detectors’ detection rates for that walker, is `r17/MECHANICS_R17.txt`. The onset rule, the three-frame endpoint and the expectations were set after this check, and the $\text{AR}(1)$ coefficient is fitted on the same file as the test.

Table S19: Pedestrian primary onsets after zeroing: 74 onsets in seven blocks. All results are descriptive. Detection within three frames, event mean [block-bootstrap interval] at design 10^{-2} ; intervals are understated with seven blocks.

Statistic	Rate ratio 10^{-2}	P_d [interval]	P_d at 10^{-3}
IN-ARCP	0.96	0.297 [0.165, 0.441]	0.172
Power clutter map	0.98	0.274 [0.138, 0.418]	0.149
OS clutter map	0.96	0.231 [0.105, 0.372]	0.128
Range CA-CFAR	1.10	0.074 [0.018, 0.163]	0.024

The descriptive IN-ARCP/clutter-map difference is 0.024 [0.007, 0.047]. The raw log’s apparent superiority does not override failed E1. Differences against OS and range CA are 0.067 [0.046, 0.091] and 0.223 [0.131, 0.349]. The high/low static-to-noise halves contain 37 onsets each, with differences 0.047 and 0.000 (high-minus-low descriptive interval [0.008, 0.096]). Without zeroing, three-frame detection is 0.162 for IN-ARCP and 0.149 for the clutter map; summing sixteen chirps after zeroing gives 0.243 and 0.209. Secondary-event results, onset-frame endpoints, null floors and other variants remain in the full R17 output. None establishes independent maritime onset performance.

17 Operating regime and deployment cost

These passages interpret existing outcomes and arithmetic, not new performance measurements.

17.1 The operating regime of the per-look screen

Corollary 4 gives the visibility horizon in *looks*. What that means in practice depends on how long a look lasts, and the paper’s three radars set that three orders of magnitude apart. At $m = 16$ and $P_{\text{fa}} = 10^{-2}$ the horizon is 3.7 looks at the opposite Doppler on IPIX and negative at the clutter Doppler, so with the $\Delta = 8$ guard a strong persistent target is kept for at most $\Delta + \ell^* = 11.7$ looks there. Because ℓ^* depends on $|r|$, each row below uses that radar’s own measured value:

Radar	Look interval	Guarded horizon	Measured $ r $ at that lag
IPIX, X-band	1 ms (1 kHz)	12 ms	0.96 (median over units)
NetRAD, S-band	1 ms (1 kHz)	12 ms	0.97–0.98 (at or near the clamp)
JKU, 77 GHz	200 ms (frame)	2.2 s	0.12 (median over bins)

Correlation and target emergence jointly constrain the look interval.

Short enough that the clutter is still correlated. The whitening gain is the clutter attenuation of the one-step predictor (Proposition 2), so it follows $|r|$ at the look lag. At 1 kHz on both sea-clutter radars $|r|$ is near one, and against the same comparator — the raw-power OS detector at the first look — the measured gain is 10.62 dB [7.05, 12.69] on IPIX and censored at ≥ 12.2 dB on NetRAD, where IN-ARCP already reached $P_d = 0.5$ at the lowest SCR tested. The two are not separable. Between 77 GHz frames $|r|$ is 0.12 and the measured gain is nil: IN-ARCP detects 0.30 of the pedestrian onsets against 0.27 for a power clutter map, a difference of 0.03 on 74 onsets falling in seven blocks, which is descriptive and not a difference. Proposition 2 predicted that parity in the frozen R17 protocol, written before the pedestrian file was downloaded, although the frame-to-frame correlation behind the prediction had been measured on a companion recording of the same scenario. It is consistent with the gain being the predictor’s clutter attenuation; the pre-specified and met IPIX comparison above is the stronger evidence.

Target emergence must outpace scale contamination. Range-cell residence is the cell width divided by radial speed, but crossing a bin boundary does not create a guaranteed abrupt onset. A point target’s range response moves continuously

across bins; its sidelobes and main-lobe entry can contaminate the next bin’s history before the peak arrives. The ramp experiments show why this matters. The following arithmetic assumes an ideal abrupt onset into each clean bin at the most favorable Doppler. It gives a horizon fraction, not a measured detection duty:

Radial speed (m/s)	Looks in one cell	Ideal horizon fraction
5	3,000	0.39%
10	1,500	0.78%
30	500	2.35%
100	150	7.83%
300	50	23.48%

The ideal horizon fraction is $\Delta + \ell^*$ divided by the looks in a cell. Two caveats keep the right-hand column from being read as a specification. First, at 1 kHz the unambiguous radial-speed interval at X-band ($\lambda \approx 0.032$ m) is about ± 8 m/s, so the faster rows fold in Doppler, and the detector does not control whether the folded Doppler avoids the clutter line. Second, ℓ^* is itself Doppler-dependent and the table uses its opposite-Doppler value, which is the best case.

The design consequence is the one stated in design rule 1 of the paper: measure $|r|$ at the intended look interval before choosing it, and read the horizon in the time units of that interval. A staring radar at 1 kHz sits at one extreme, where the gain is large and onsets are rare. A scan-to-scan clutter map sits at the other, where a new return can have a longer horizon but the correlation gain may be lost. The paper’s evidence covers neither the middle nor any operational scan pattern, so no intermediate look rate can yet be recommended. Real target range responses and clutter correlation must be measured together at the proposed lag.

17.2 What a cell-wise detector costs

The paper puts the detector where a range CFAR stage would go, so that is the comparison a system designer needs.

Arithmetic. Per range cell and per pulse: p complex multiply–accumulates for the $\text{AR}(p)$ innovation, one magnitude-squared, one add and one subtract to slide the scale window, one division and one comparison. That is $O(p)$ complex and $O(1)$ real operations, the same order as a sliding-window range CFAR, and the reference-window sum also admits a constant-cost sliding update.

State. $m + \Delta + p = 25$ complex samples per range cell, for $m = 16$, $\Delta = 8$ and $\text{AR}(1)$. A range CFAR needs no per-cell slow-time state at all; a clutter map needs one level per cell. This is the real cost of the construction, and it scales with the number of cells.

Threshold storage. One scalar per statistic. Because innovation normalization makes the score close to pivotal — in one analysis at 10^{-2} , texture-quintile spread 1.7 against 87.5 for the unnormalized statistic and 9.8 for normalization by the raw history RMS — one threshold can serve every range cell, and a cell’s level need not be learned or stored. That is the operational payoff of the pivotality result. It is only approximate: the same spread is 2.7 at 10^{-4} on IPIX and 2.40 on NetRAD, so one shared threshold does not hold the per-cell rate exactly at the rates the paper cares about.

Clean data for calibration. Section IV-F gives $\mathbb{P}\{P_{\text{fa}} > 2\alpha\} \leq 0.05$ for i.i.d. continuous scores and every n beyond about $3.15/\alpha$. At $\alpha = 10^{-4}$ that is 31,477 episodes. An episode spans $m + \Delta + 1 = 25$ pulses once design rule 1’s guard is in place, and the clutter cells are the 14 range cells less the target cells and one guard cell on each side, which is 7 to 9 per IPIX file, not 14. Drawn as non-overlapping episodes and pooled over those cells, it is 87 to 112 s of clean recording at 1 kHz. This is an i.i.d. planning calculation, not a guarantee for temporally or spatially dependent episodes. Non-overlap does not ensure independence; drift and dependence require separate validation when the regime changes.

18 Complete frozen outputs and protocols

Saved summary outputs and machine-readable plotting inputs are available in the repository. Raw recordings and per-episode NPZ intermediates are not bundled. The original intermediate arrays are retained in the local study archive; reproducing them requires the provider data and study code. Failed conditions, ambiguous attribution, censored endpoints and superseded log values remain accessible. The two corrections in Section 7 accompany reuse. The PDF retains every low-rate Part-L ratio and interval, but does not transcribe every per-SCR grid, order, clip setting or diagnostic variant. These extra numerical entries were not independently recomputed in this revision.

Repository base: <https://github.com/razaumair2203-ux/inarc/tree/r21-trs-submission/research/r7c>.

- Main protocol: `PROTOCOL.md`; detection/onset protocols: `detection/PROTOCOL_DETECTION.md`, `PROTOCOL_ONSET.md`; baselines: `baselines/PROTOCOL_BASELINES.md`; reference radar: `jku/PROTOCOL_JKU.md`.
- R10–R15 and R17: `r10/PROTOCOL_R10.md` through `r15/PROTOCOL_R15.md` for the studies reported here (R10, R11, R12, R14, R15), and `r17/PROTOCOL_R17.md`; verdicts are the corresponding `RESULTS.R*.md`. NetRAD deviations: `r15/DEVIATIONS_R15.md`; pedestrian mechanics check: `r17/MECHANICS_R17.txt`.

- r10/: audit_existing.txt; na4_convergence.txt; r10_summary.txt; theory_ramps.txt.
- baselines/: baselines_summary.txt.
- detection/: detection_summary.txt; dwell_summary.txt; onset_summary.txt; onset_theory.json; theory_checks_output.txt.
- jku/: jku_summary.txt.
- r11/: r11_summary.txt; theory_os_output.txt.
- r12/: r12_jku_summary.txt; r12_summary.txt; verify_laws_output.txt.
- r14/: r14_summary.txt; theory_check_guard_output.txt; theory_check_independent_output.txt.
- r15/: explore_r15_attribution_output.txt; explore_r15_flag_extent_output.txt; explore_r15_hotspots_output.txt; r15_summary.txt.
- study/results/r17/: r17_summary.txt.
- Corrected calibration sizes: r12/calibration_size.txt.

References

- [1] P. J. Brockwell and R. A. Davis, *Time Series: Theory and Methods*, 2nd ed. New York, NY, USA: Springer, 1991.
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